

NX-414: Brain-like computation and intelligence

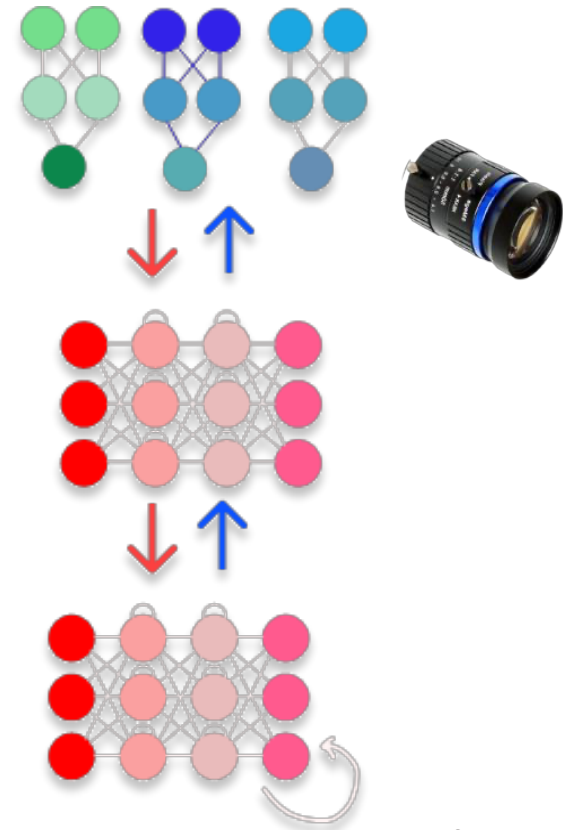
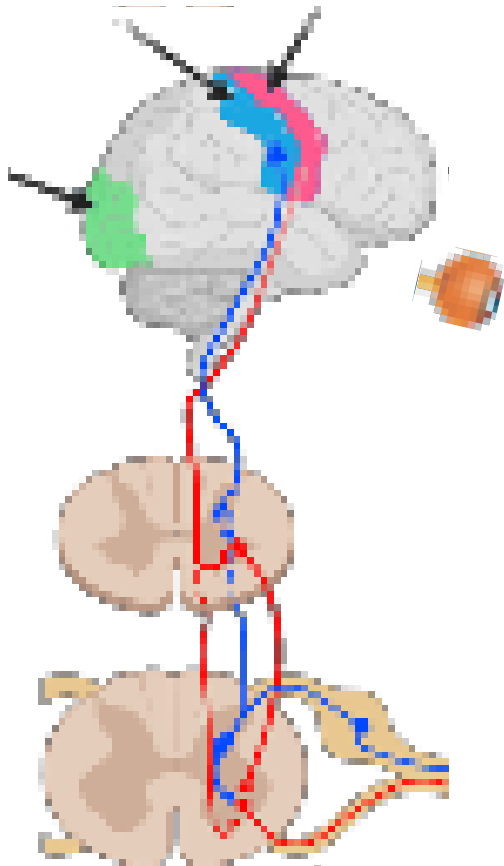
Martin Schrimpf

Lecture 5, March 19 2025

Biological Intelligence



Artificial Intelligence



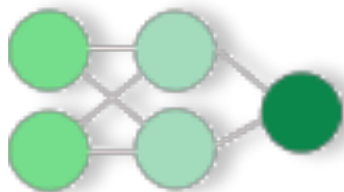
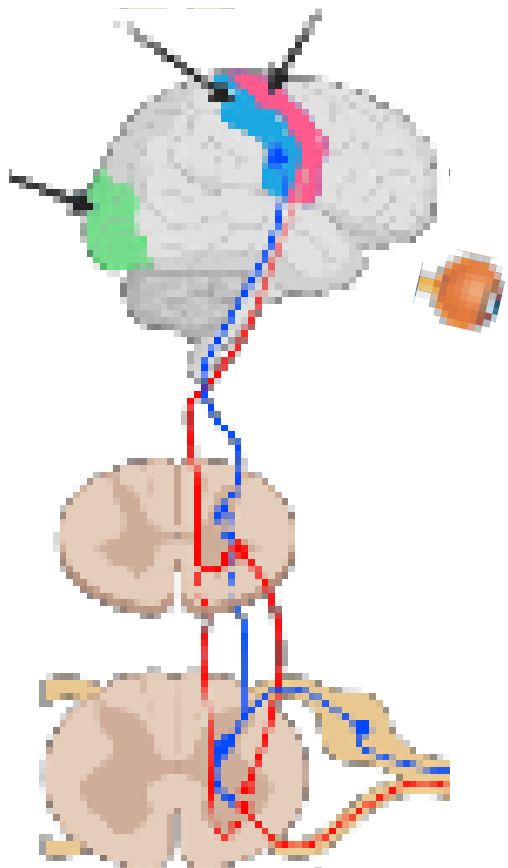
Normative frameworks

Information theoretic

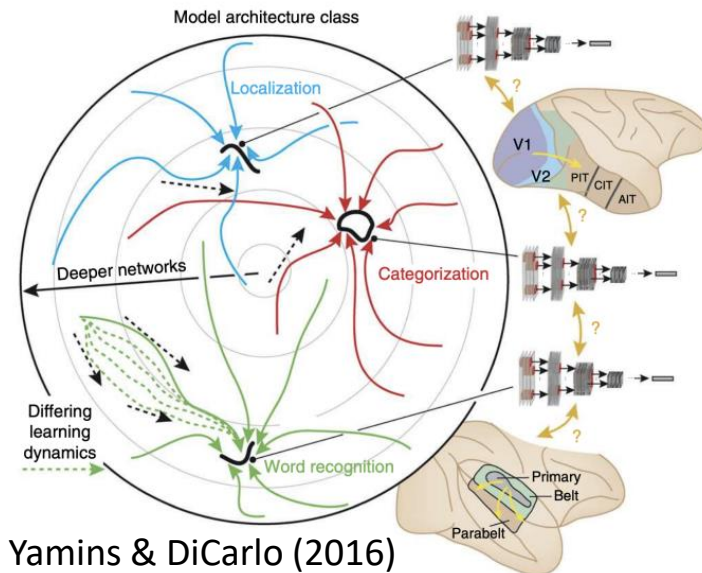
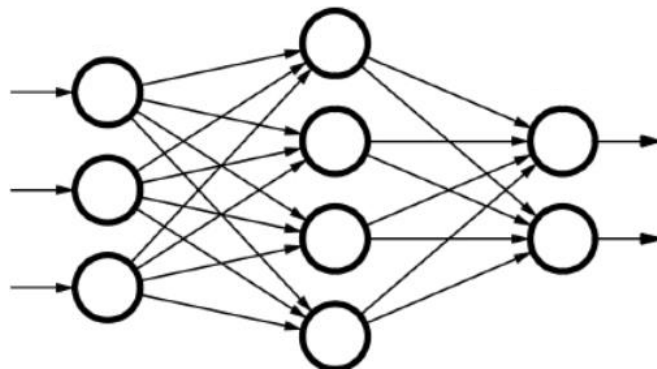
e.g. sparse coding,
redundancy reduction,
mutual information ...

Utilitarian

e.g. **recognize objects**,
chase prey, navigate ...



Using deep neural networks as goal-driven models of a system



Vision: object recognition.

Yamins & Hong et al. (2014), Schrimpf & Kubilius et al. (2018)



Audition: speech recognition, speaker & sound identification. Kell et al. (2018)



Somatosensation: shape recognition.
Zhuang et al. (2017)



Language: next-word prediction.
Schrimpf et al. (2021)

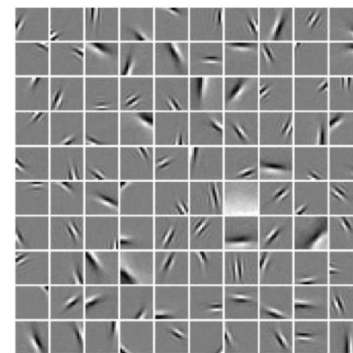
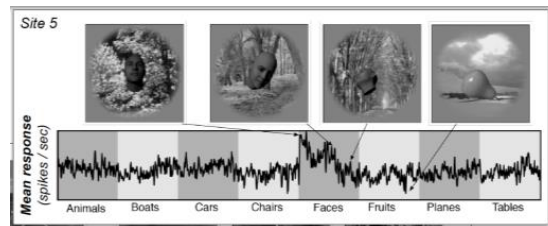


Decision making: context-dependent choice. Mante & Sussilo et al. (2013)

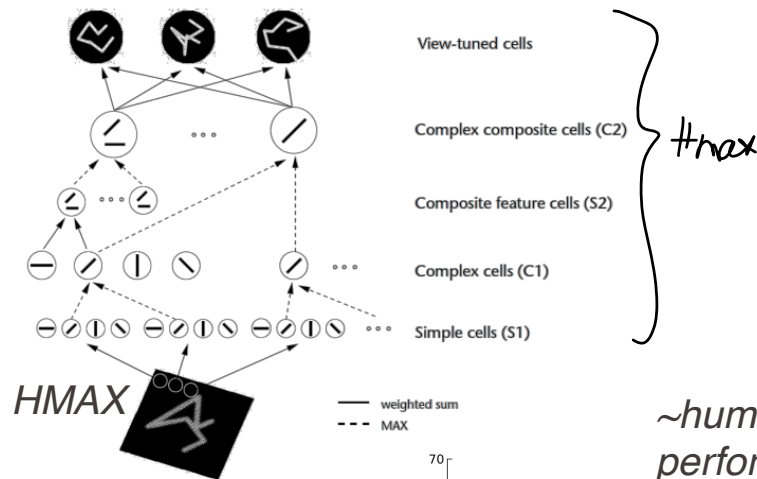
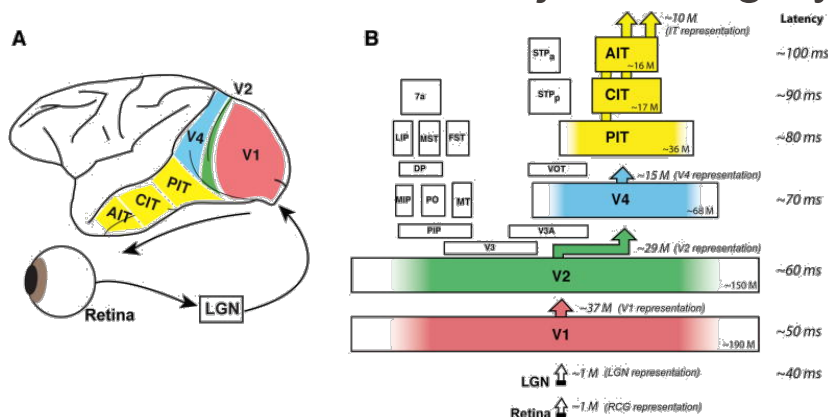


Proprioception: action recognition.
Sandbrink et al. (2023)

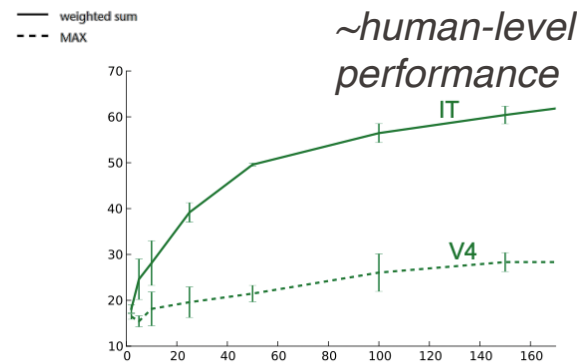
- Difficult for a long time to scale models beyond simple responses in V1



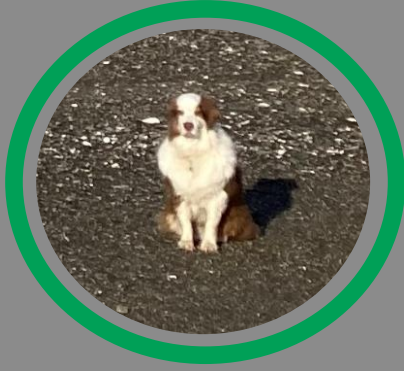
- IT –linear-readout → object category



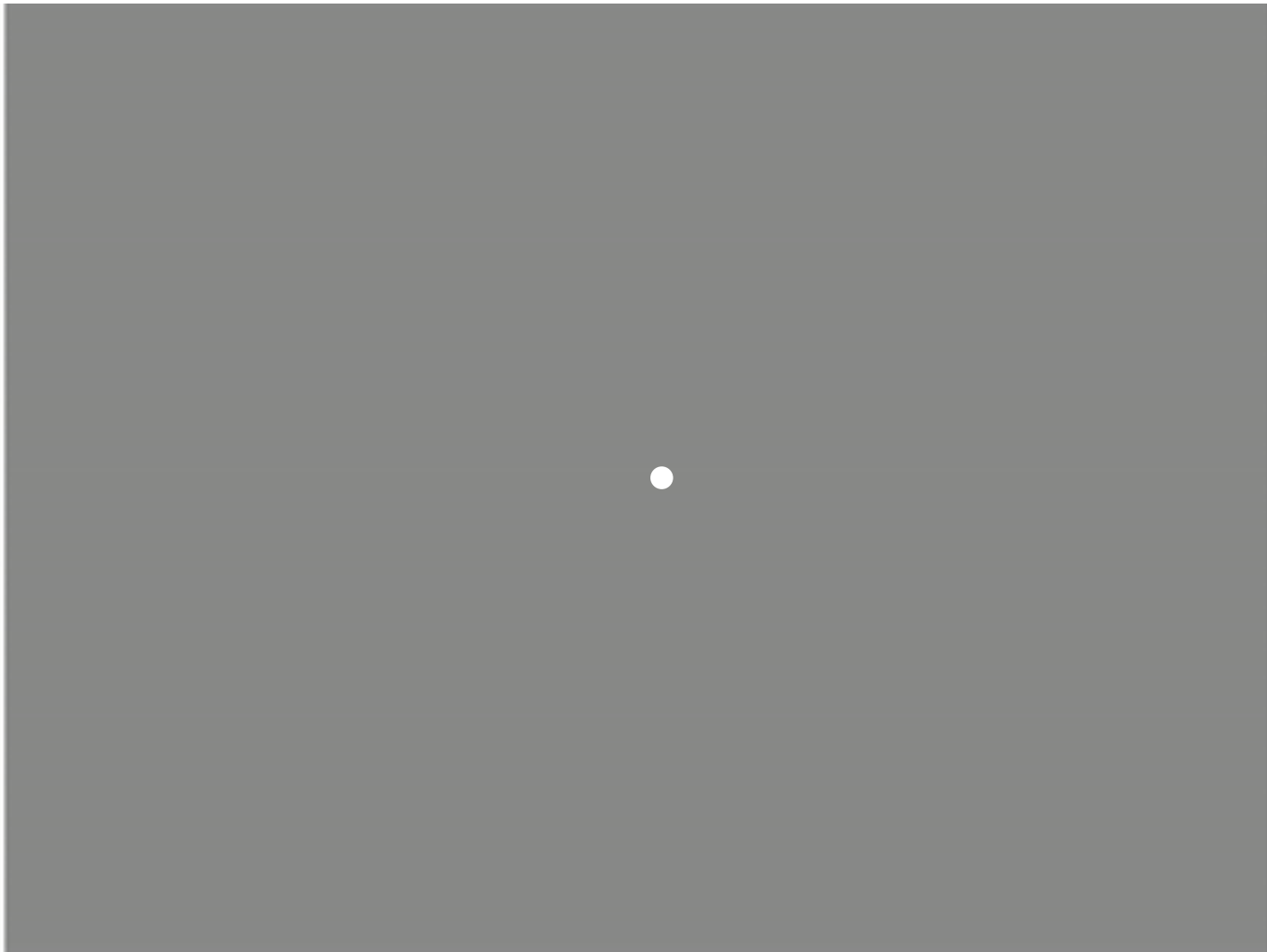
- Task optimization → brain-like internal representations

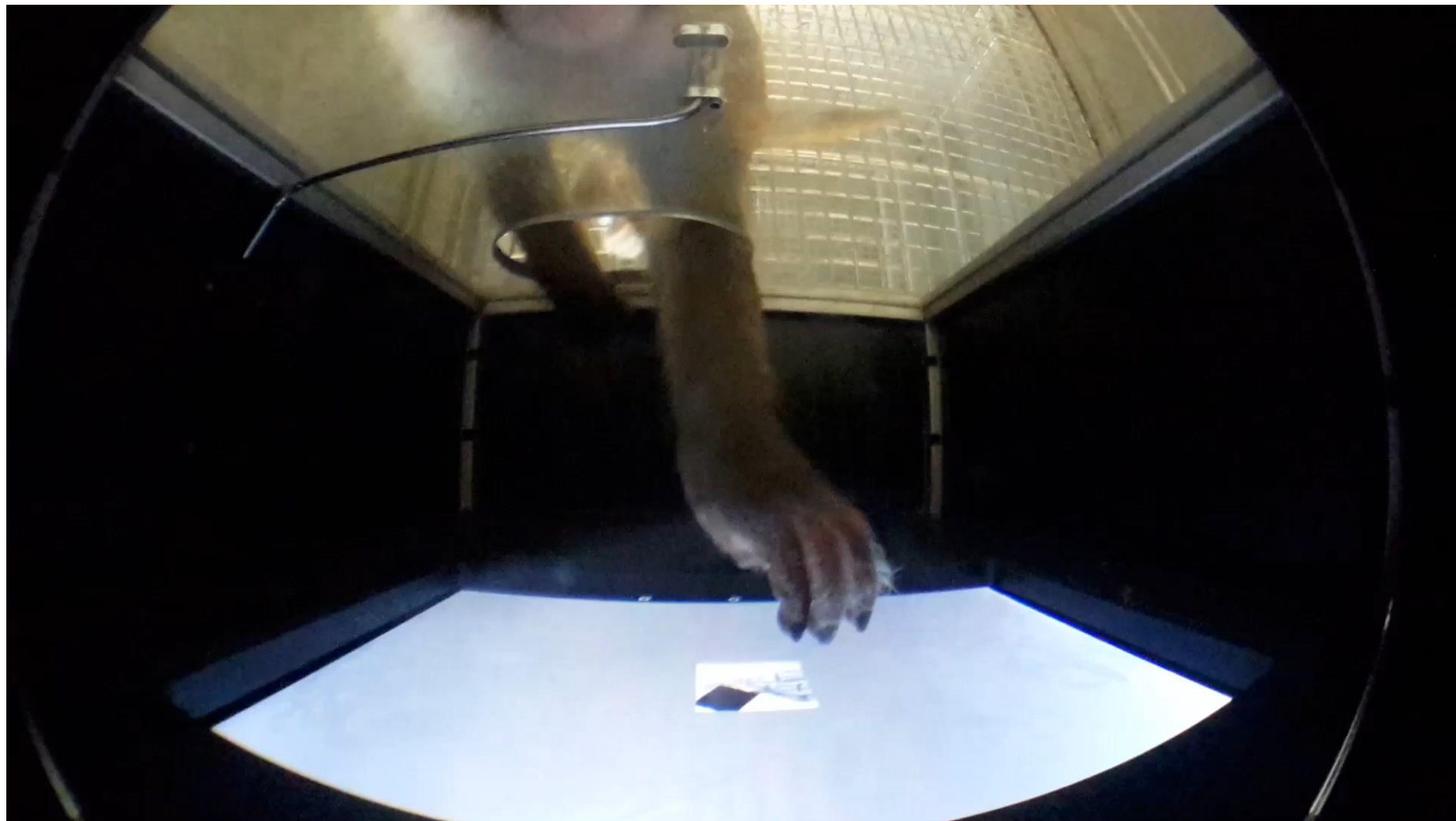


Behavioral experiment

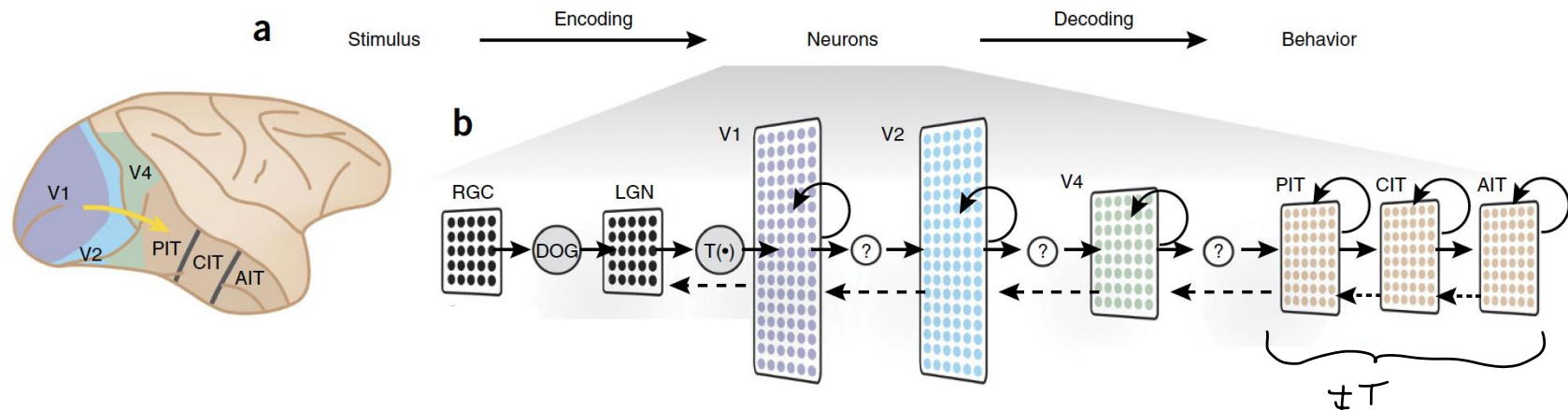


Behavioral experiment



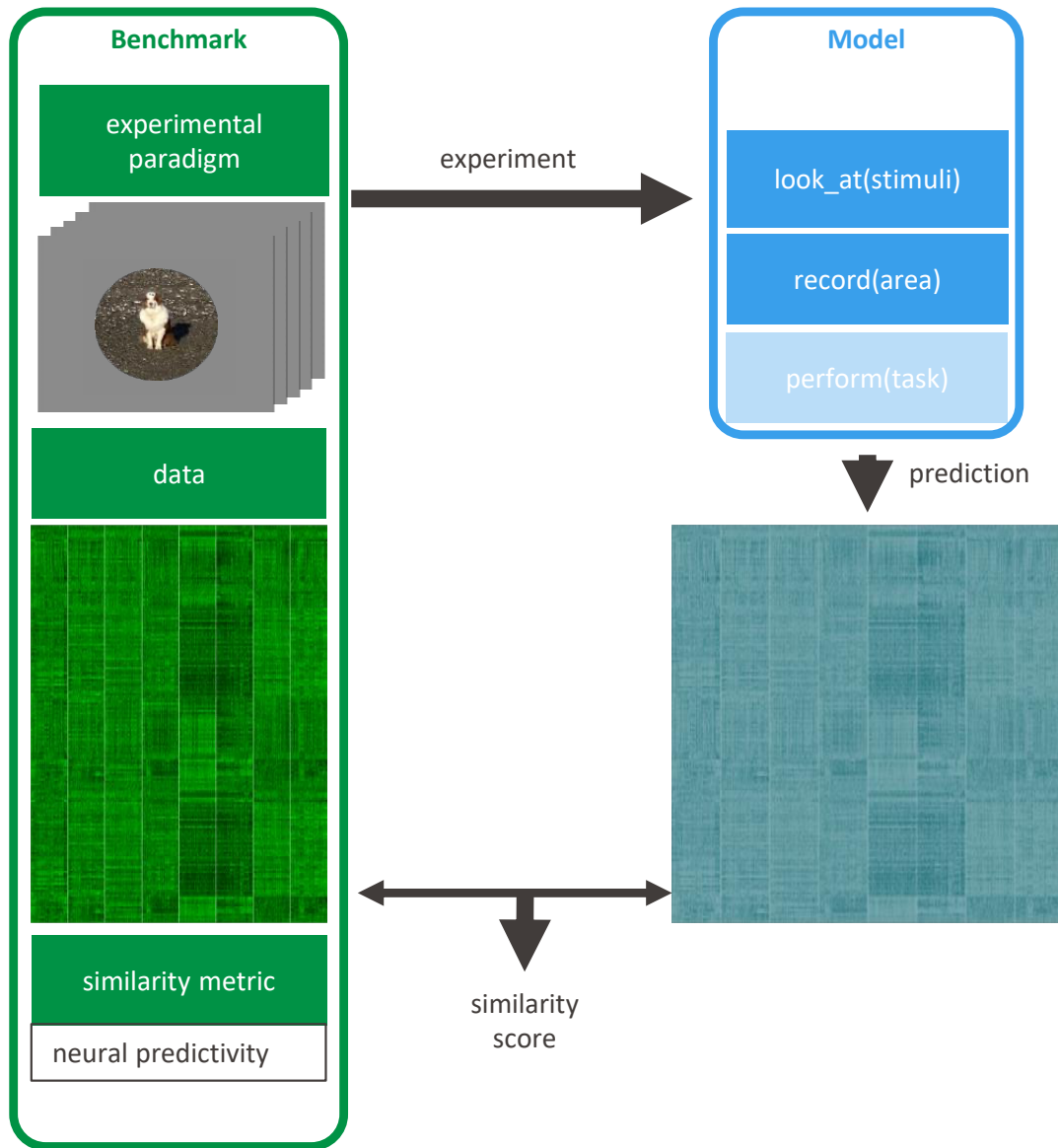


Core-object recognition and the visual pathway



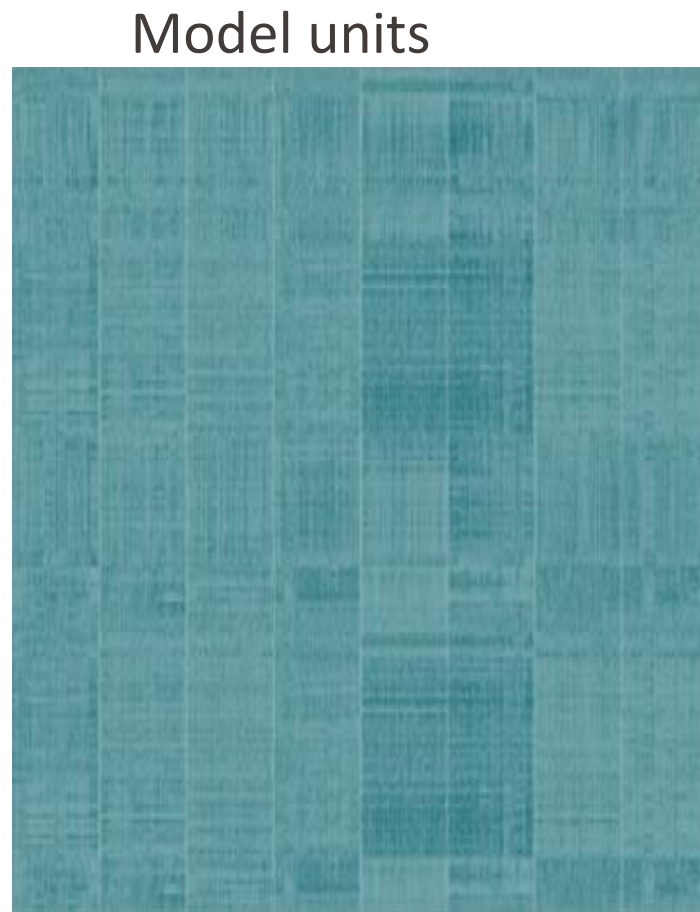
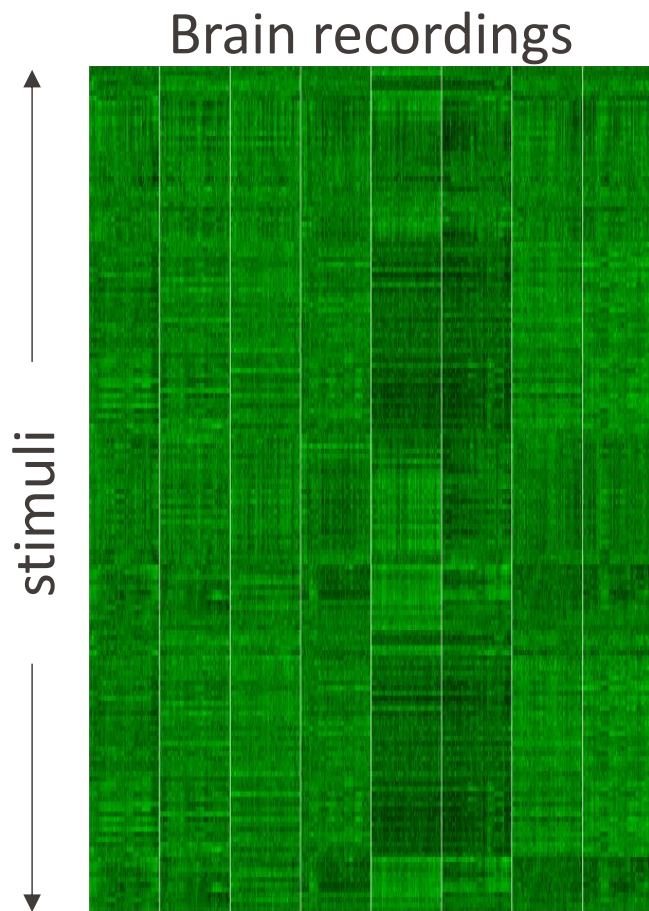


Neural benchmark



Neural alignment = alignment between
stimulus-matched recordings

Neural benchmarks



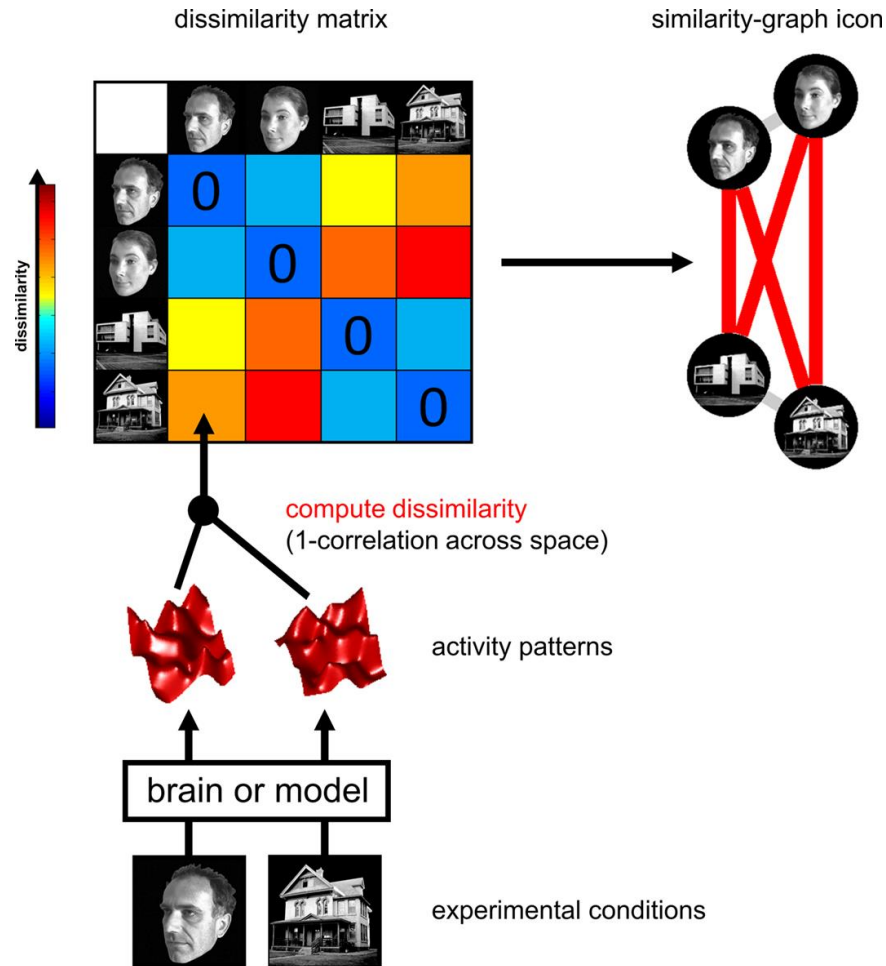
Representational similarity analysis (RSA)

-> Way to study similarity between model prediction and biological data

$$RDM(\phi(x))_{ij} = 1 - \frac{cov(\phi(x_i), \phi(x_j))}{\sqrt{var(\phi(x_i)) \cdot var(\phi(x_j))}}$$

-> How similar the system responses are

—> comparing the system's (either brain or model) response to different stimuli. Why ? See next slide



similarity metric

Representational similarity analysis (RSA)

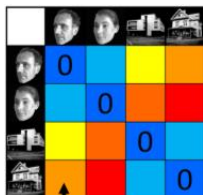
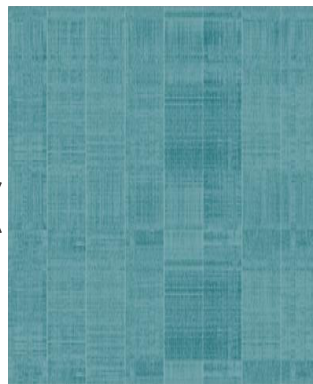
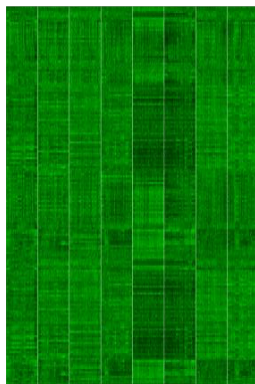
Brain recordings

Model units

$$\text{score} = \text{corr}(\text{RDM}(\text{Brain recordings}), \text{RDM}(\text{Model units}))$$

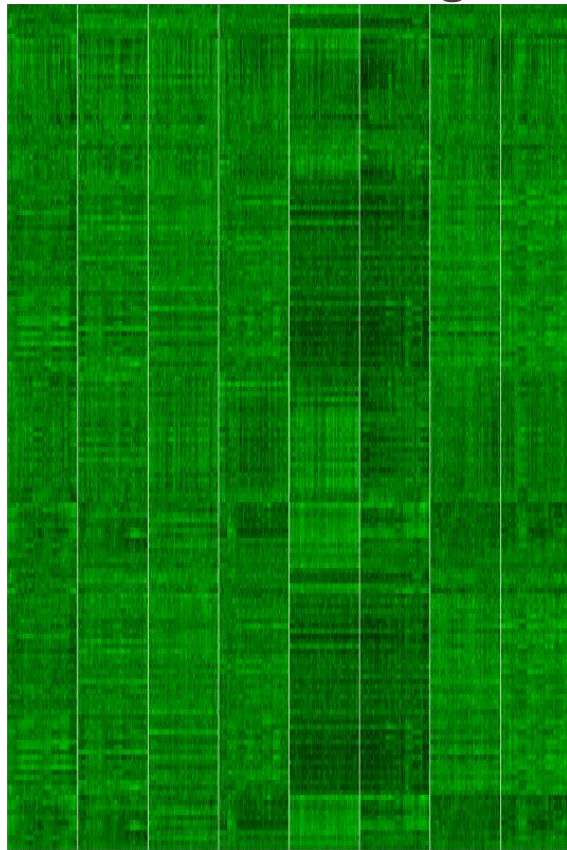
*correlate upper
triangular only*

(Because symmetric)



Neural benchmarks

stimuli

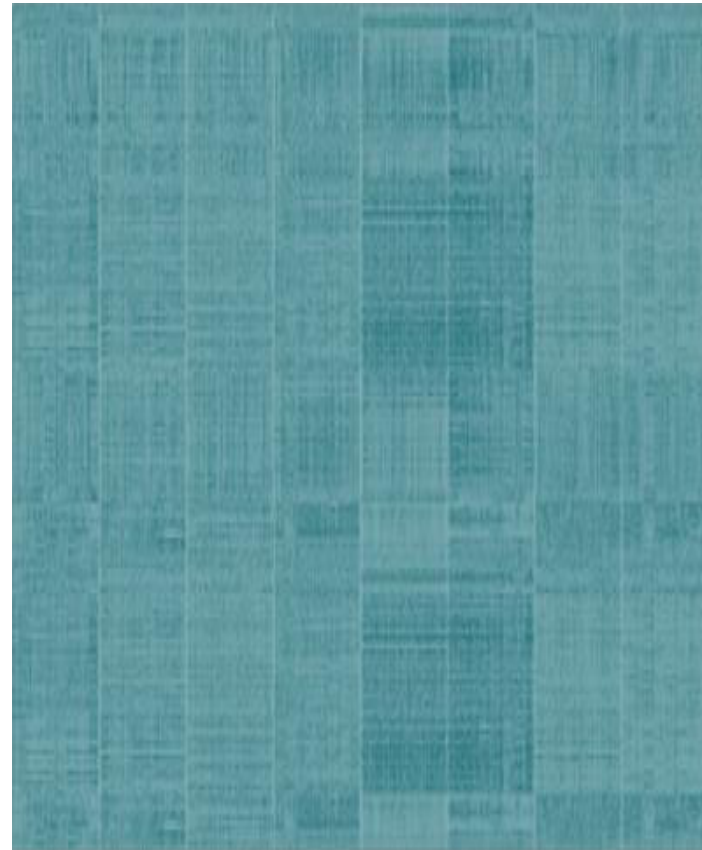


similarity metric

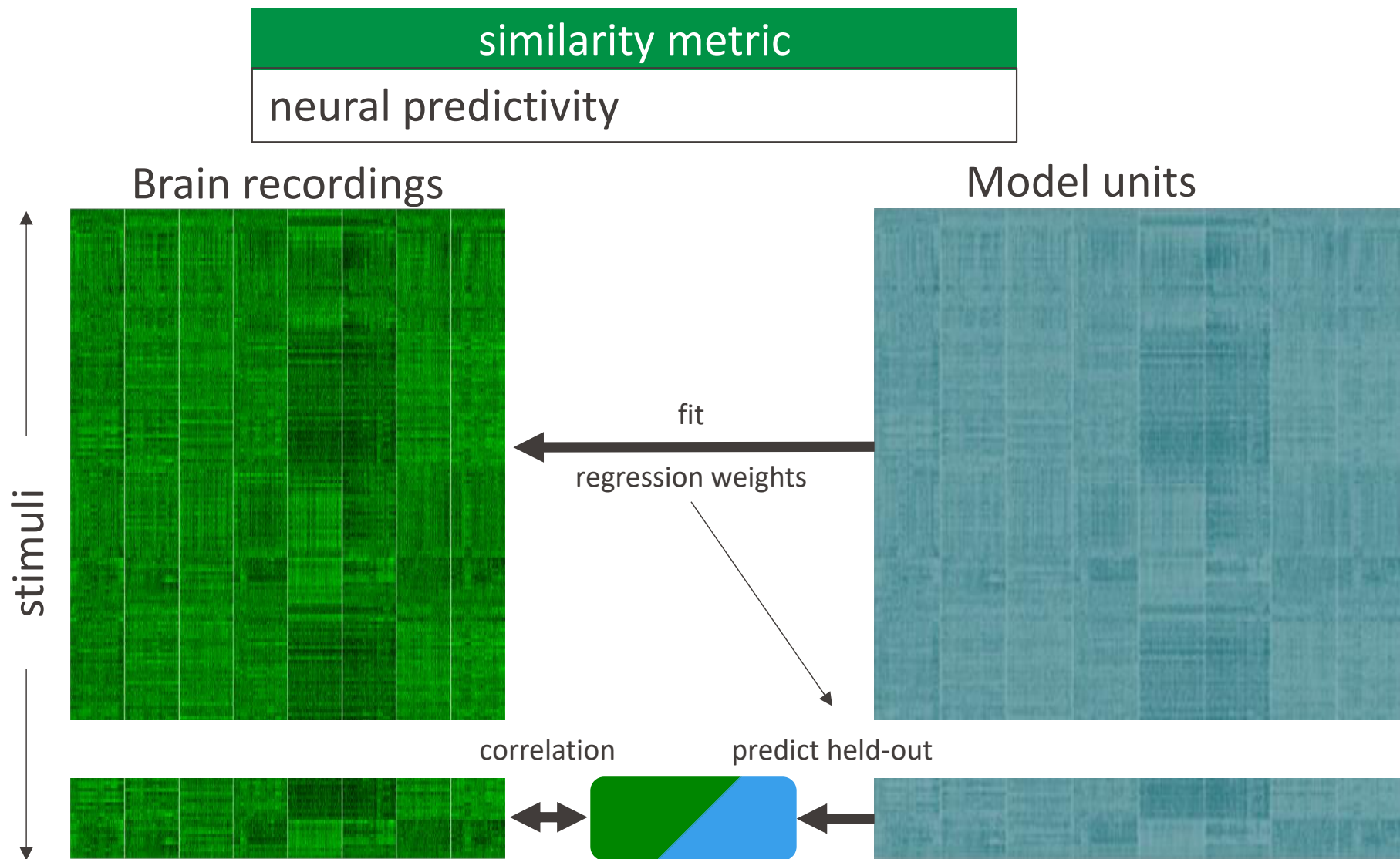
neural predictivity

Can we find a linear regression from model activity to brain activity such that it predicts our data?

Model units

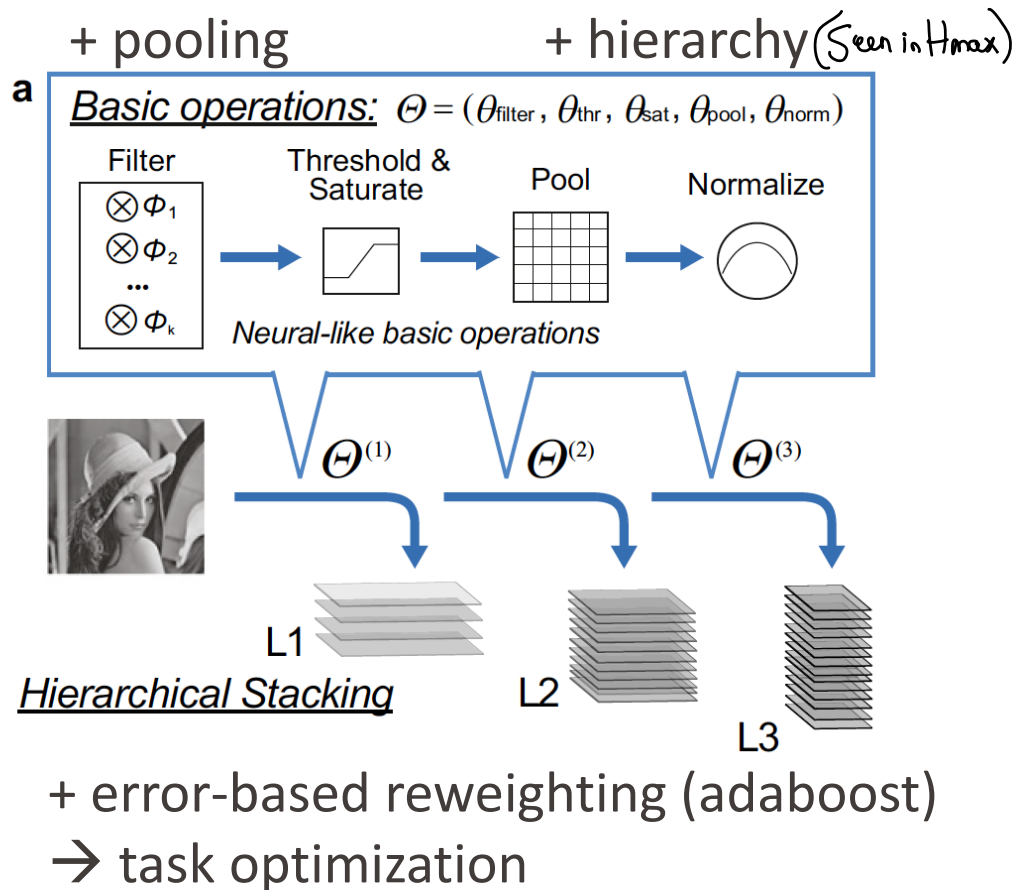
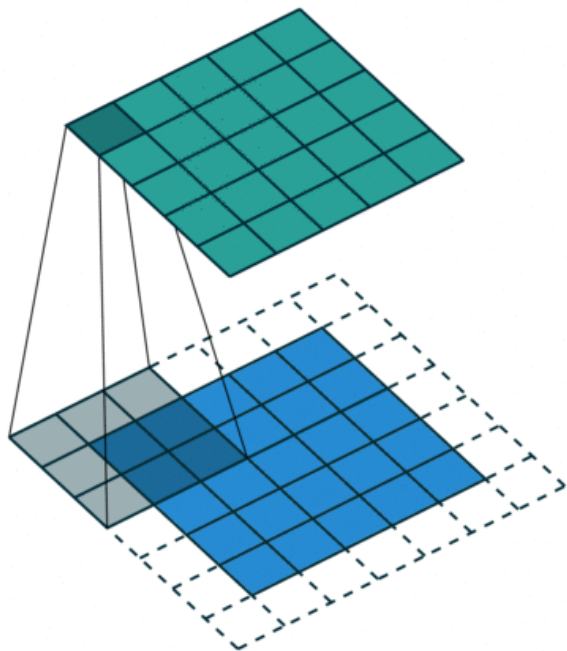


Neural benchmarks





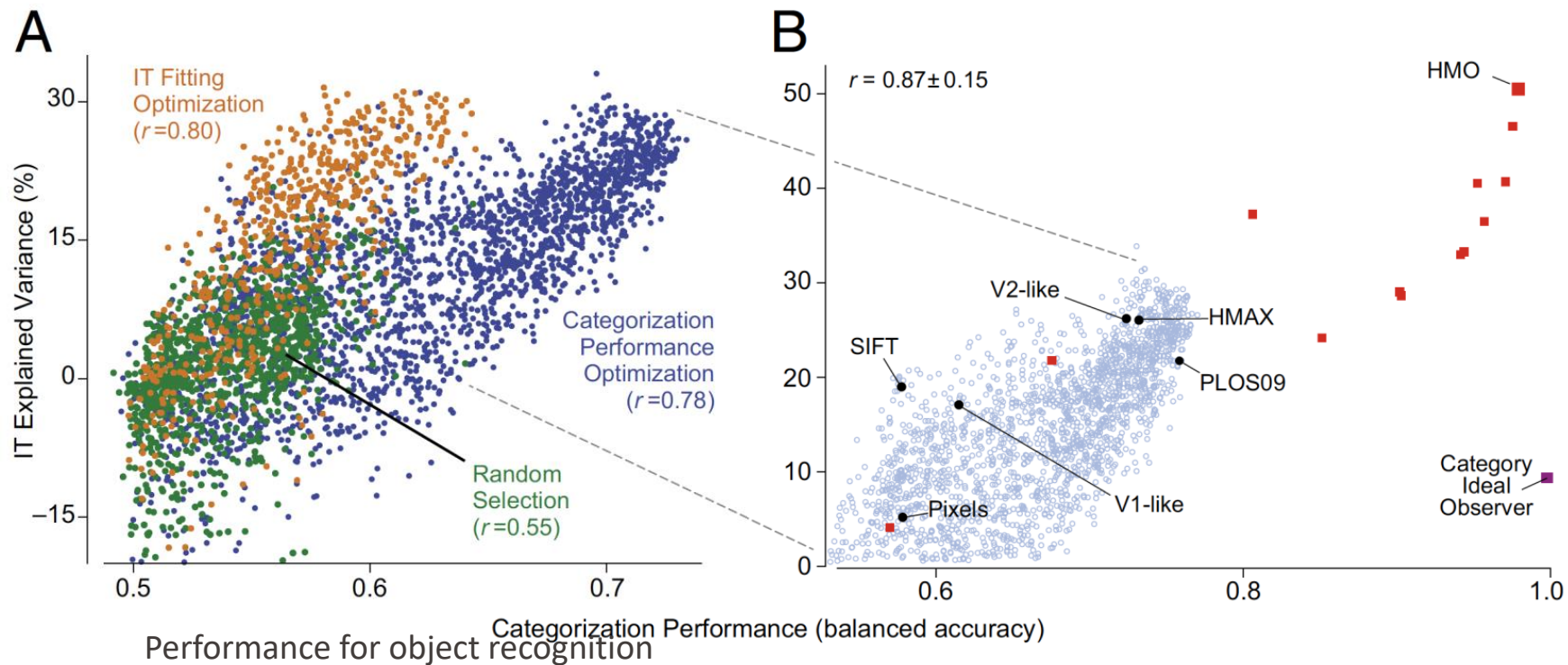
Convolutions



via lots of compute

(Pinto et al. 2009; Bergstra et al. 2013)

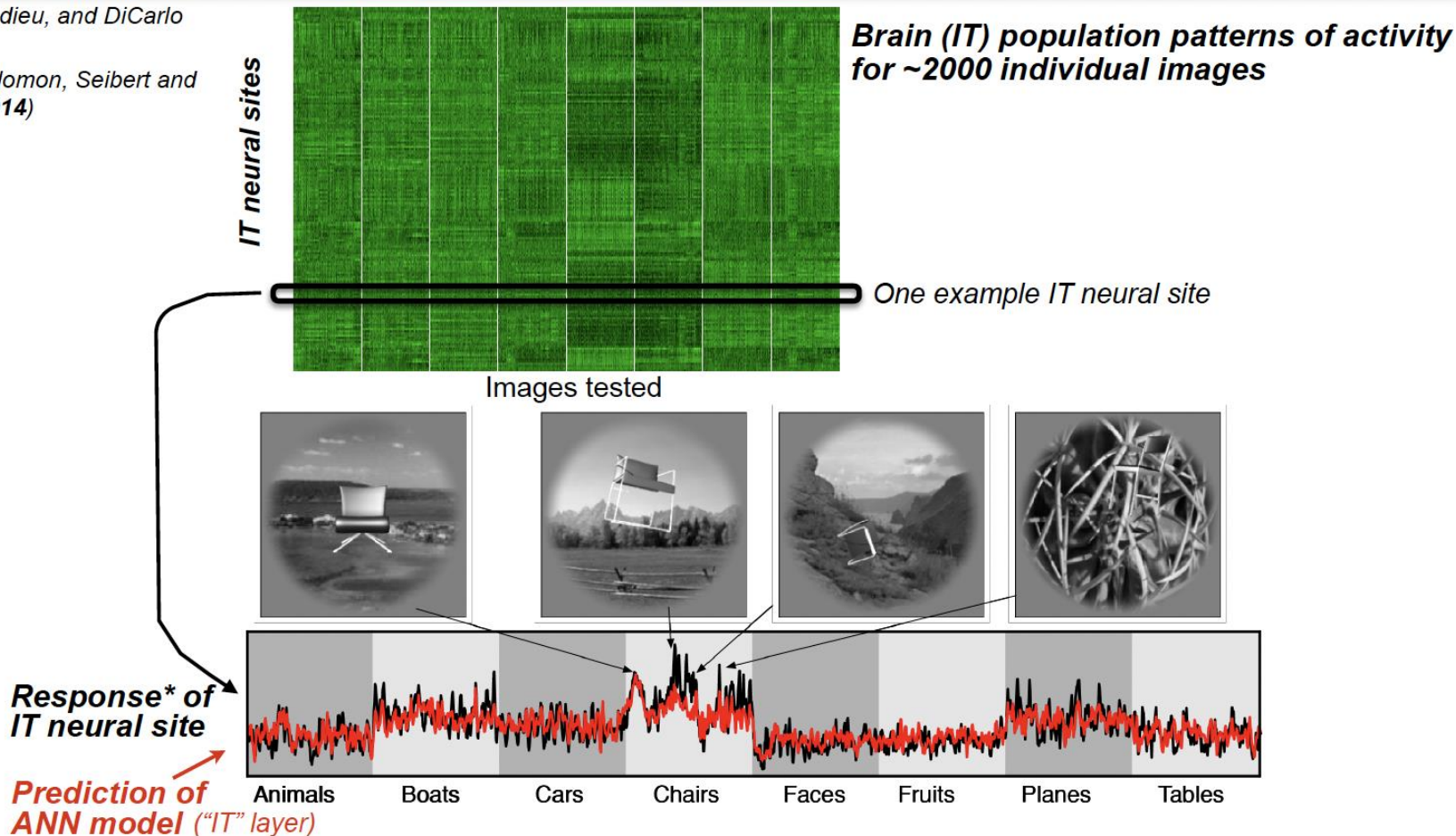
Large scale architecture search and model comparison



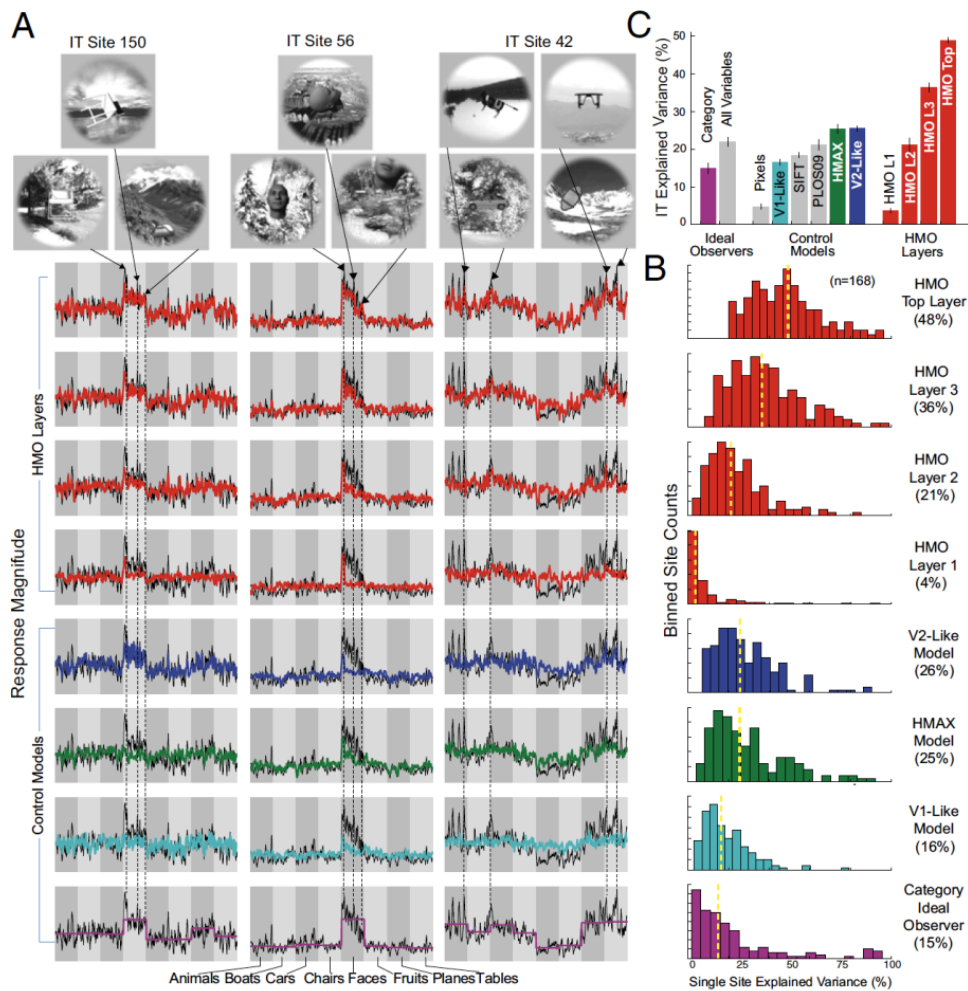
Task-trained models match brain's "hidden neurons"

Yamins, Hong, Cadieu, and DiCarlo
NIPS (2013)

Yamins, Hong, Solomon, Seibert and
DiCarlo **PNAS (2014)**



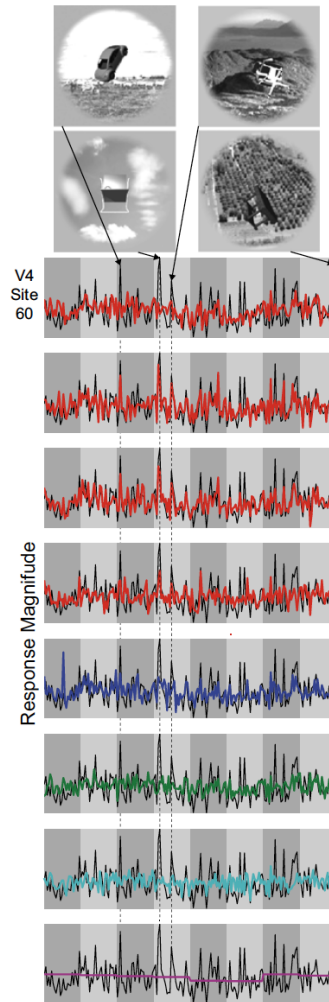
Predictions for IT



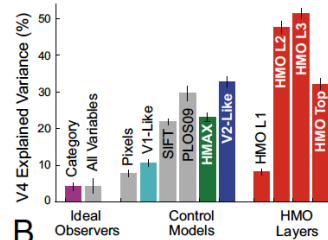
HMO top model at the time (not the case anymore apparently)

Predictions for V4

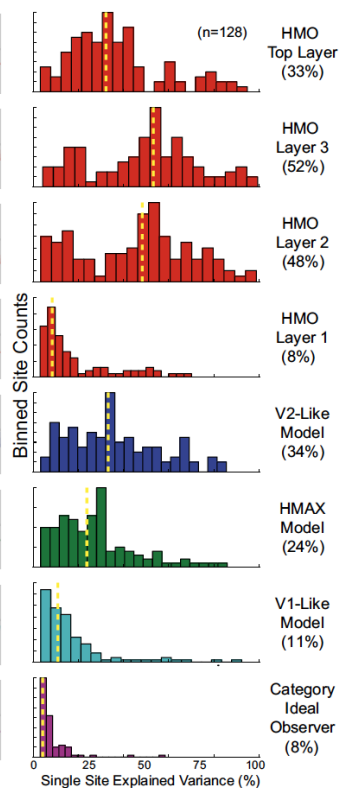
A



C

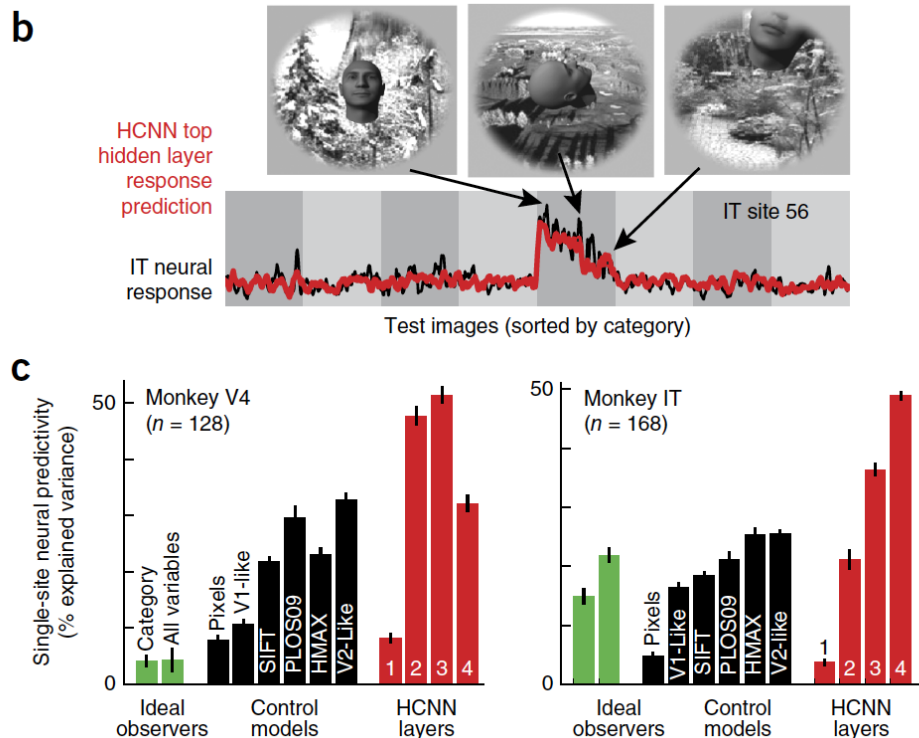


B



No longer the top layer that is best, but the one before

Hierarchy as a consequence of task-optimization

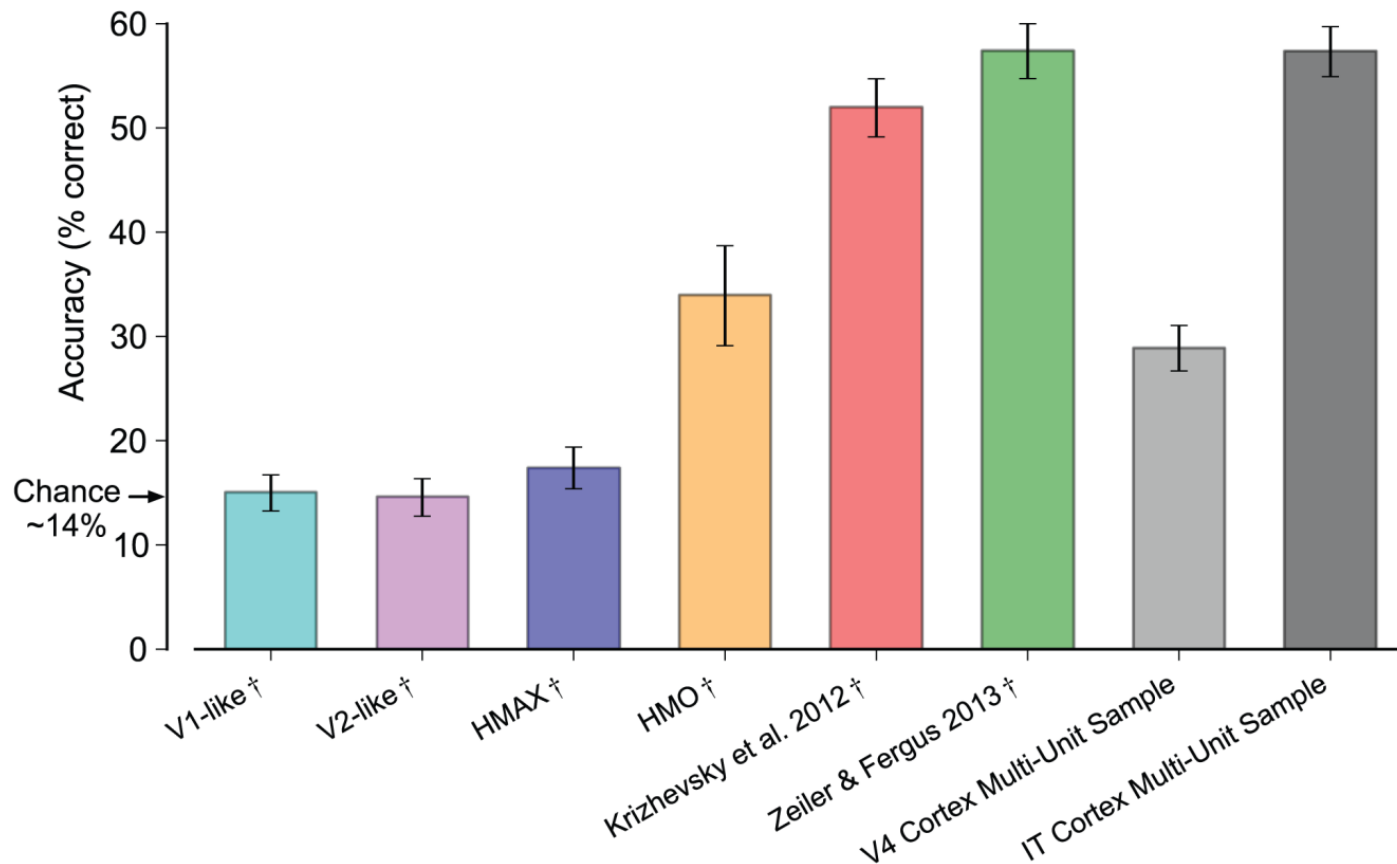


Second-to-last layer is better for V4, which kinda makes sense because V4 is the second-to-last layer of the ventral stream

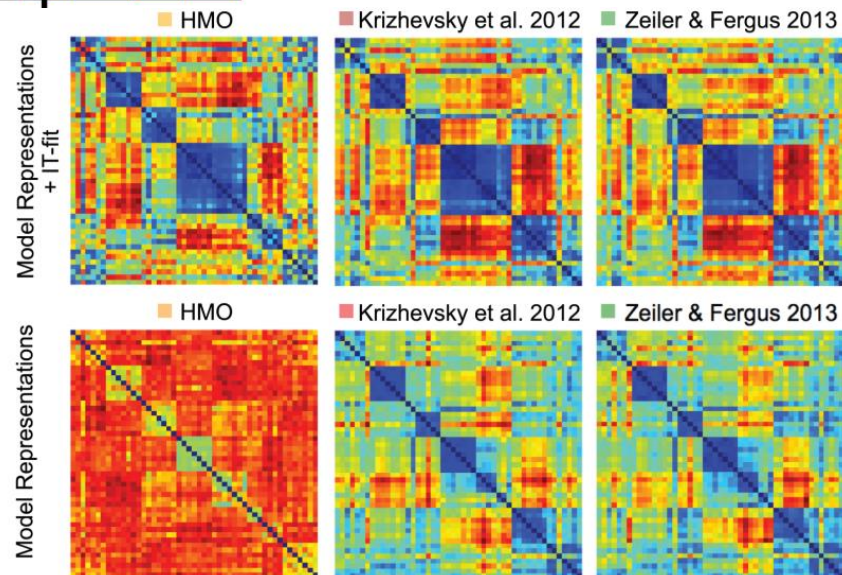
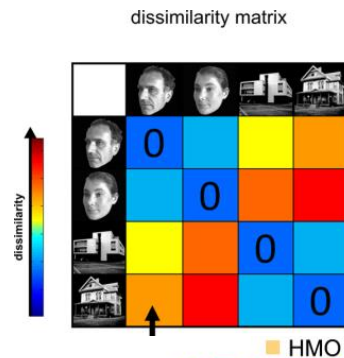
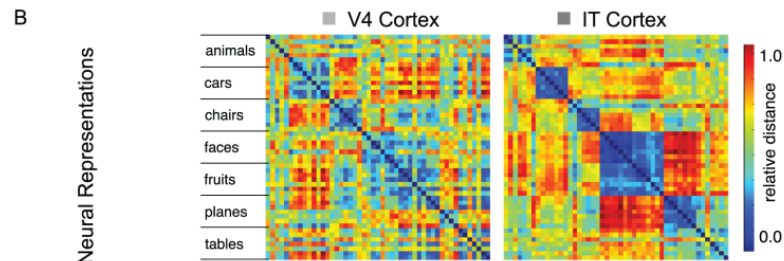
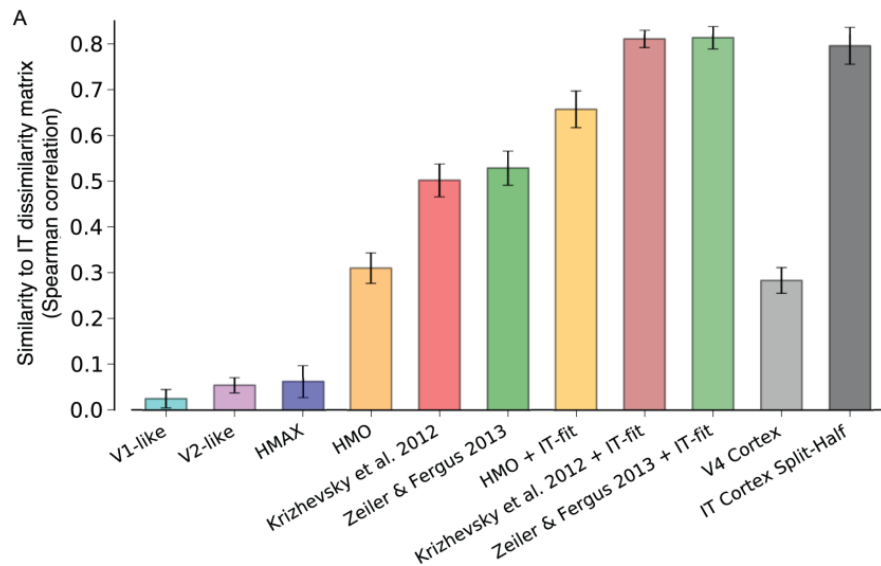
Last layer better for IT, which kinda makes sense because IT is last layer

Category classification

Comparing to backprop-trained deep nets...



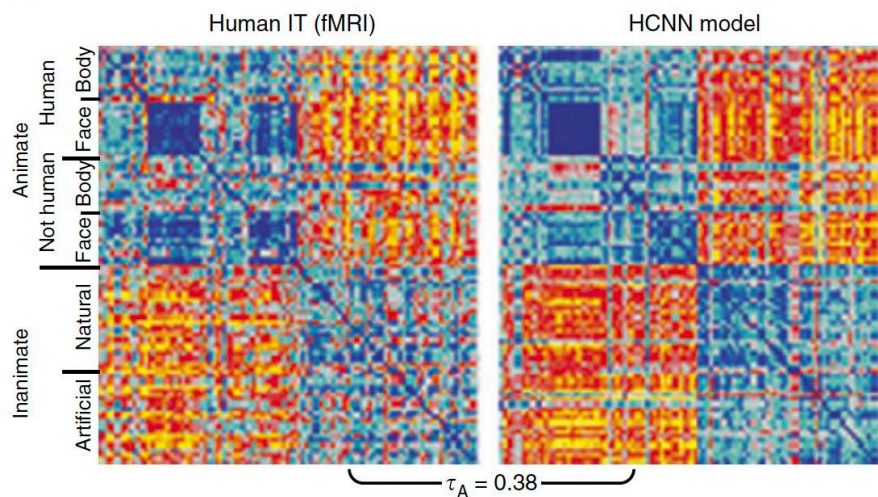
Model comparison



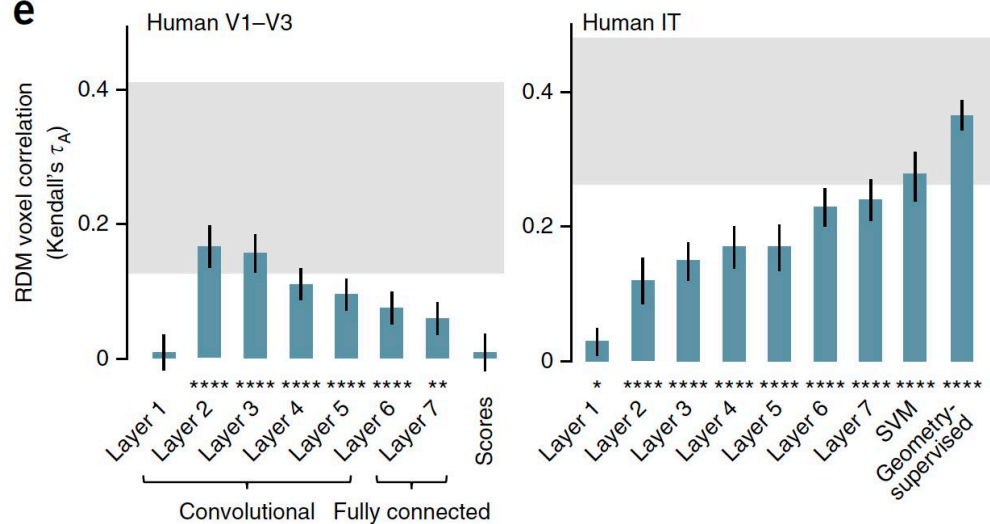
Backprop on ImageNet is much better than HMO.
Computer Vision – “as a side effect” – revolutionized modeling primate vision!

fMRI based analysis in humans

d



e



Lemanic Life Sciences Hackathon

- Work on 12 exciting projects
- Develop new skills
- Collaborate with like-minded people
- Get expert mentorship
- Get inspired by talks from Alumni
- Pitch projects and win prizes
- Have fun during your Easter break

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LIFE SCIENCES
HACKATHON**
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April 23 - 25, 2025 🕒
Hall SV and SV 1717 📍

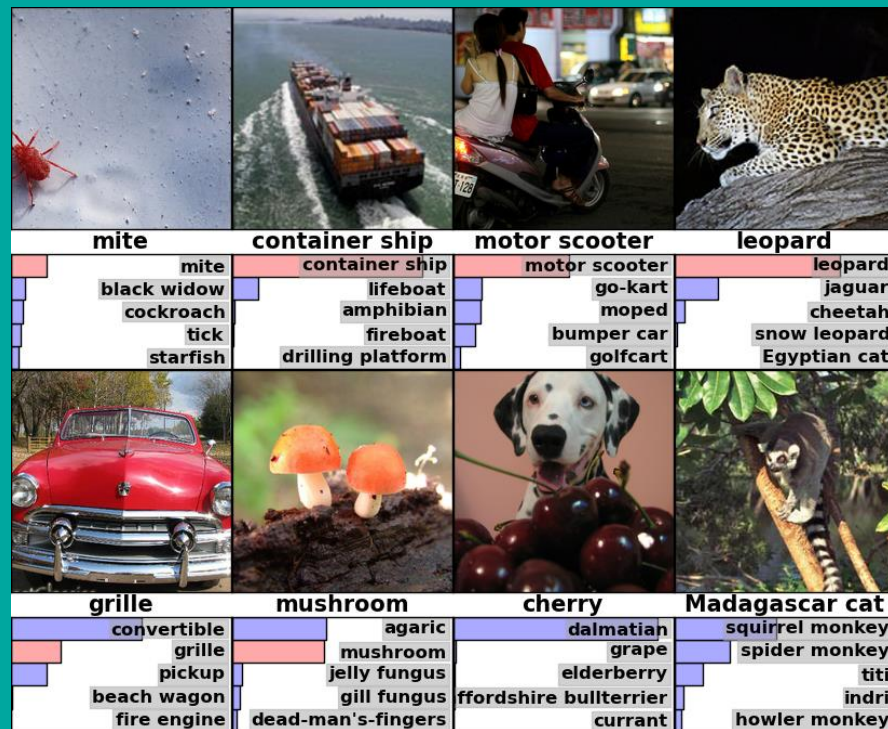
ImageNet challenge

1000 classes

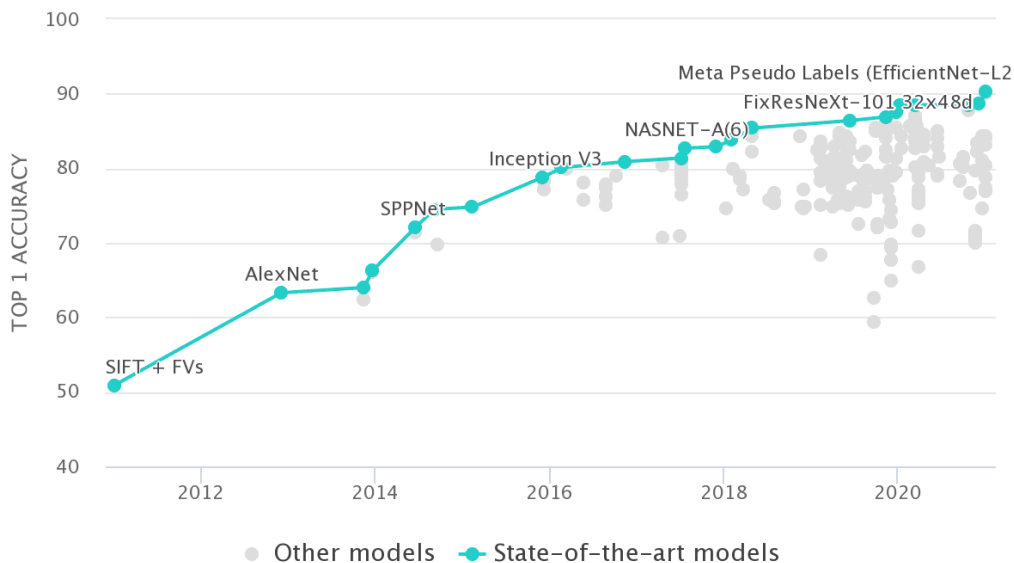
1.2 training images

150,000 test images

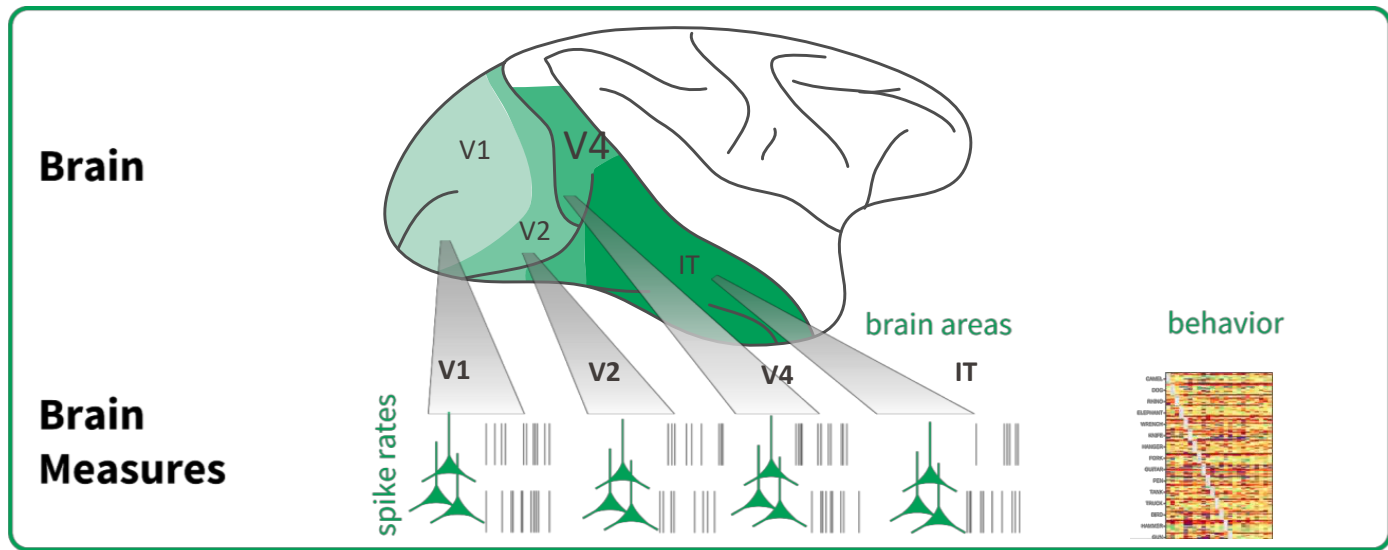
AlexNet predictions



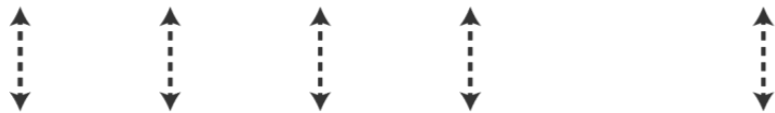
Krizhevsky, Sutskever, Hinton, NeurIPS 2012



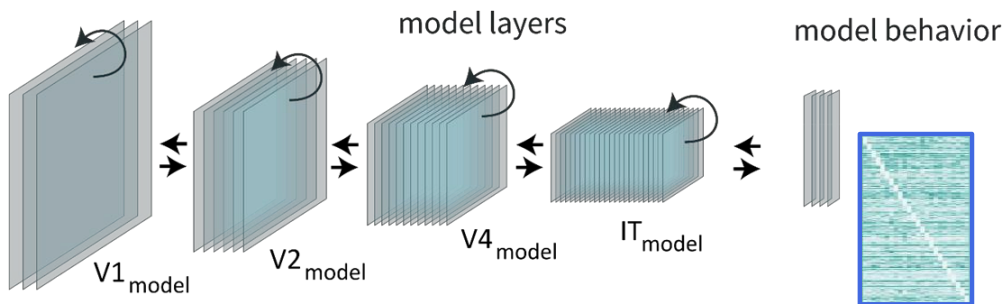
EPFL Integrative testing of models on brain + behavioral data



Test alignment



Model Candidates



Artificial neural network models

- Trained for computational task, weights optimized via backprop



- Internal processing stages (hidden layers, “deep” learning)
- Accept any new input (pixels)

Test models' alignment across visual ventral stream



V1

V2

V4

IT

behavior

Brain-Score component Score

.4
.3
.2.4
.3
.2.6
.5
.4
.3
.2.6
.5
.4
.3
.2.6
.5
.4
.3
.2
.1
.0

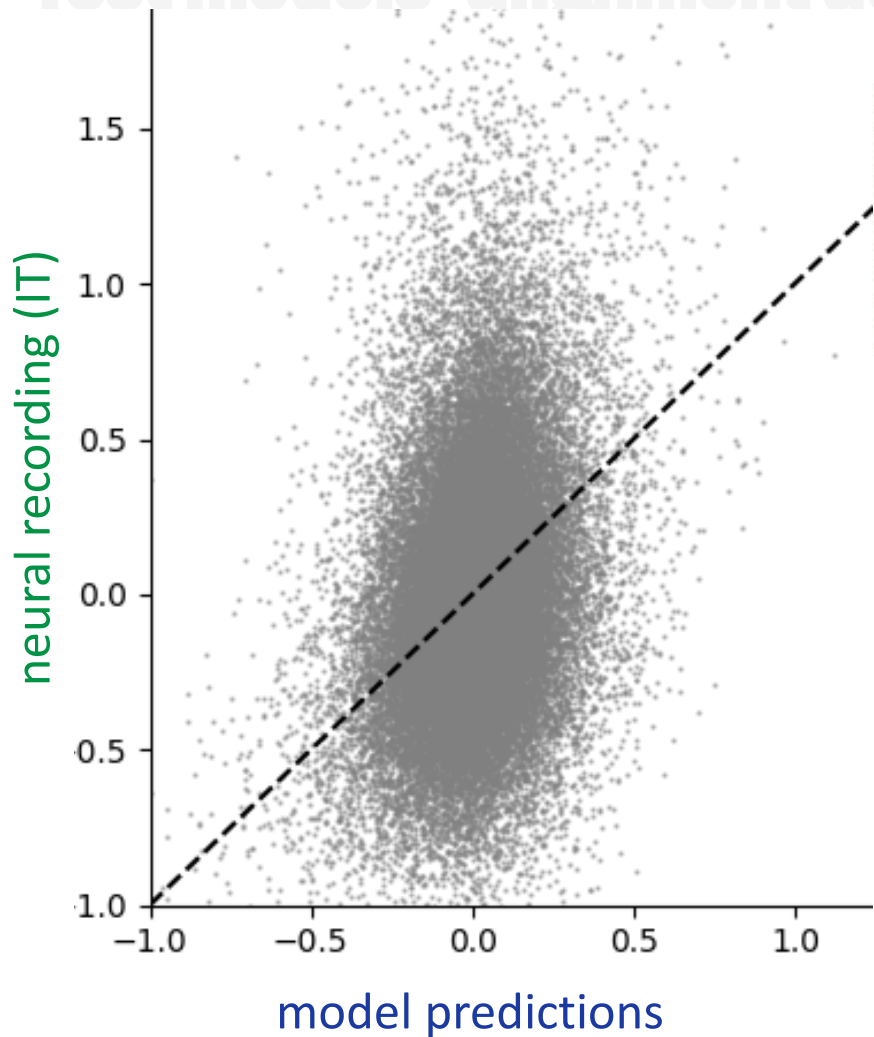
Model Score $\rightarrow$ linear predictivity
score (brain alignment)

Schrimpf, Kubilius*, et al. (2018)*

V1, V2 data: Freeman, Ziemba*, et al. (2013)*

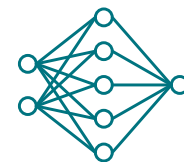
V4, IT data: Majaj, Hong*, et al. (2015)*

behavioral data: Rajalingham, Issa*, et al. (2018)*



Model candidates tested:

hmax *classic neuroscience model*



Schrimpf, Kubilius*, et al. (2018)*

V1, V2 data: Freeman, Ziemba*, et al. (2013)*

V4, IT data: Majaj, Hong*, et al. (2015)*

behavioral data: Rajalingham, Issa*, et al. (2018)*

Test models' alignment across visual ventral stream

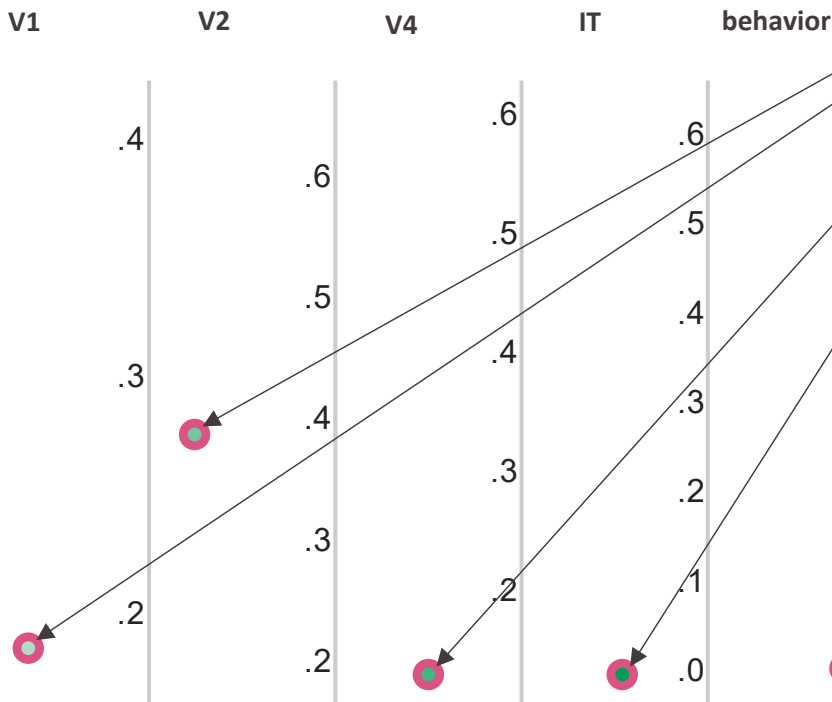


Model candidates tested:



hmax *classic neuroscience model*

Brain-Score component Score



Schrimpf, Kubilius*, et al. (2018)*

V1, V2 data: Freeman, Ziemba*, et al. (2013)*

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Test models' alignment across visual ventral stream



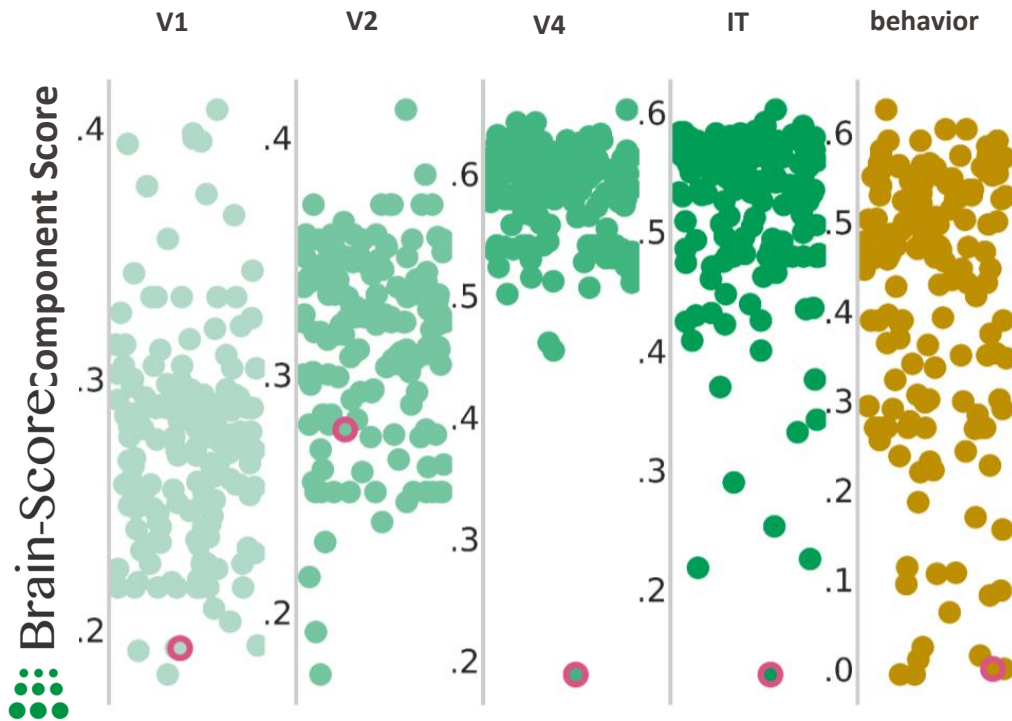
Model candidates tested:



hmax
vgg-16
vgg-19
densenet-121
densenet-169
densenet-201
inception_resnet_v2
inception_v1
inception_v2
inception_v3
inception_v4
mobilenet_v1_0.25_128
mobilenet_v1_0.25_160
mobilenet_...
mobilenet_v2_1.3_224
mobilenet_v2_1.4_224

ML models

nasnet_large
nasnet_mobile
pnasnet_large
resnet-101_v1
resnet-101_v2
resnet-152_v1
resnet-152_v2
resnet-18
resnet-34
resnet-50_v1
resnet-50_v2
squeezeNet1_0
squeezeNet1_1
xception
...



Schrimpf, Kubilius*, et al. (2018)*

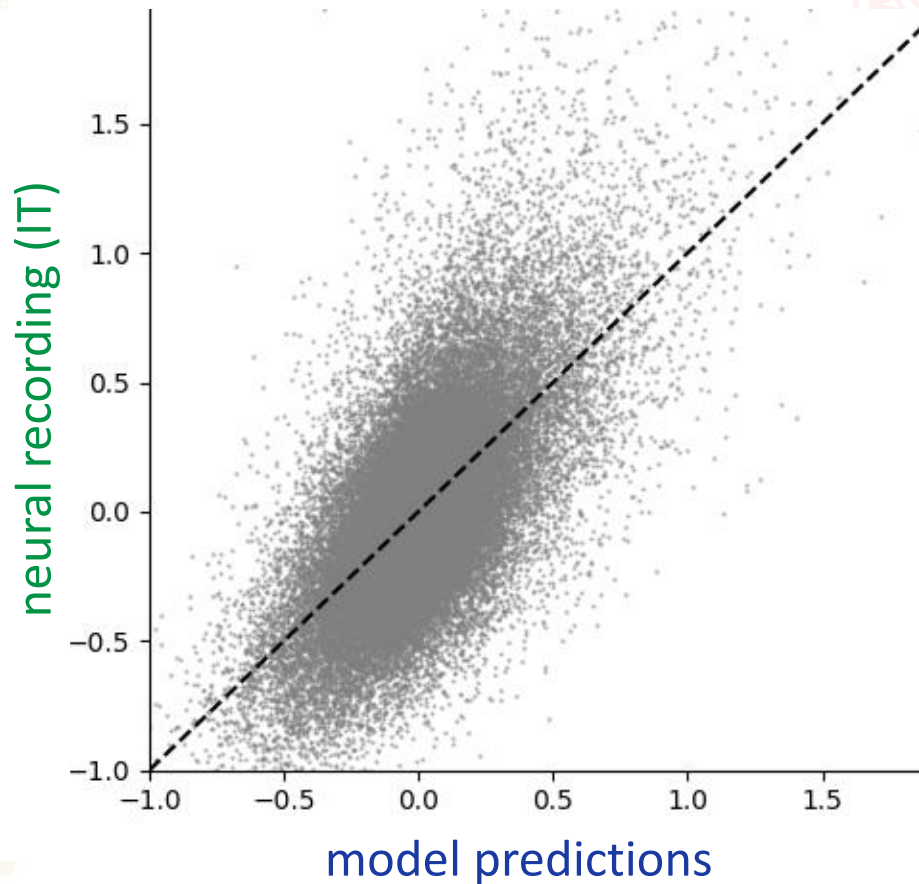
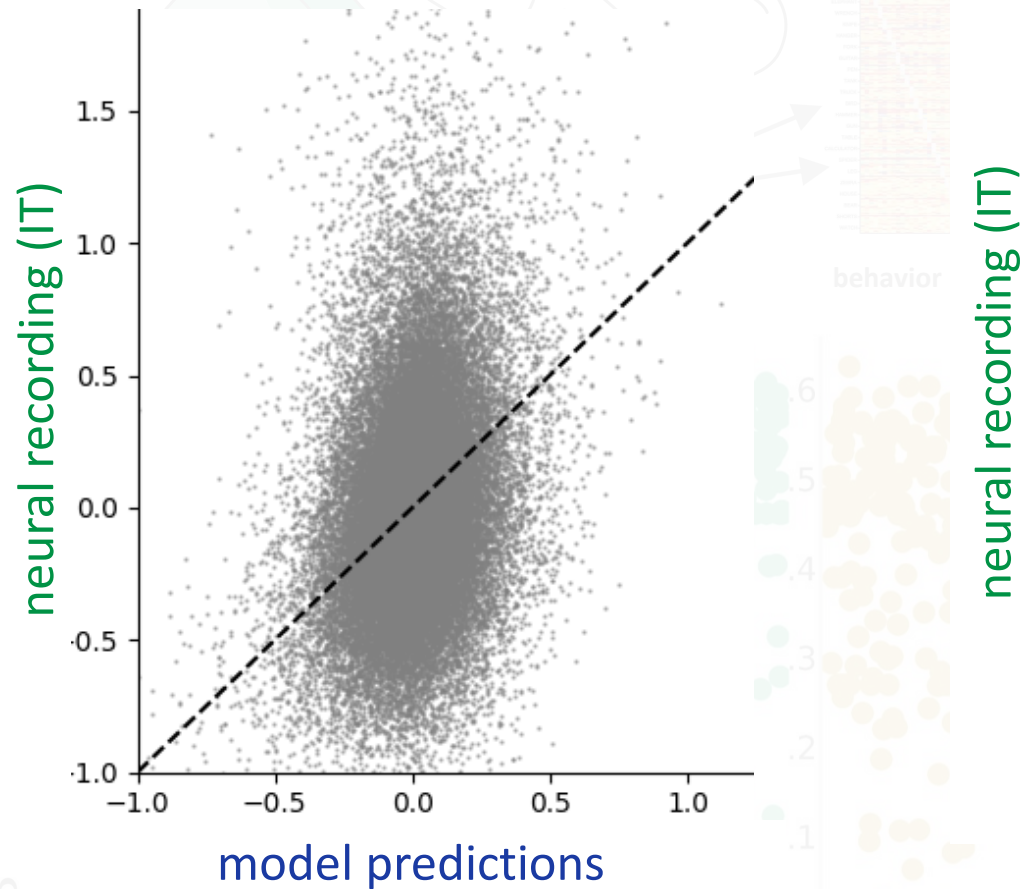
V1, V2 data: Freeman, Ziemba*, et al. (2013)*

V4, IT data: Majaj, Hong*, et al. (2015)*

behavioral data: Rajalingham, Issa*, et al. (2018)*

hmax

resnext-wsl



Test models' alignment across visual ventral stream



V1 V2 V4 IT behavior

Brain-Score component Score

Certain artificial neural networks are **state-of-the-art models of neural and behavioral alignment** (not perfect)

Model candidates tested:

hmax
vgg-16
vgg-19
densenet-121
densenet-169
inception-v3
inception-v4
inception-resnet-v2
mobilenet-v2
mobilenet-v2_1.3_224
mobilenet-v2_1.4_224

nasnet_large
nasnet_mobile
pnasnet_large
resnet-101_v1



All Computer Vision models are **trained on a task** without biological data

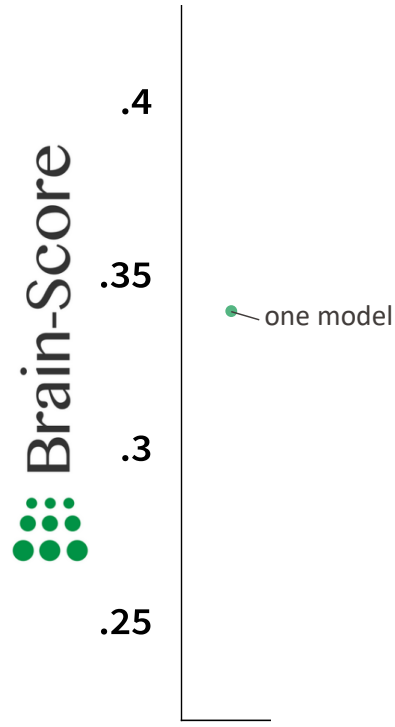
Schrimpf, Kubilius*, et al. (2018)*

V1, V2 data: *Freeman*, Ziemba*, et al. (2013)*

V4, IT data: *Majaj*, Hong*, et al. (2015)*

behavioral data: *Rajalingham*, Issa*, et al. (2018)*

EPFL What explains the model differences?



EPFL What explains the model differences?

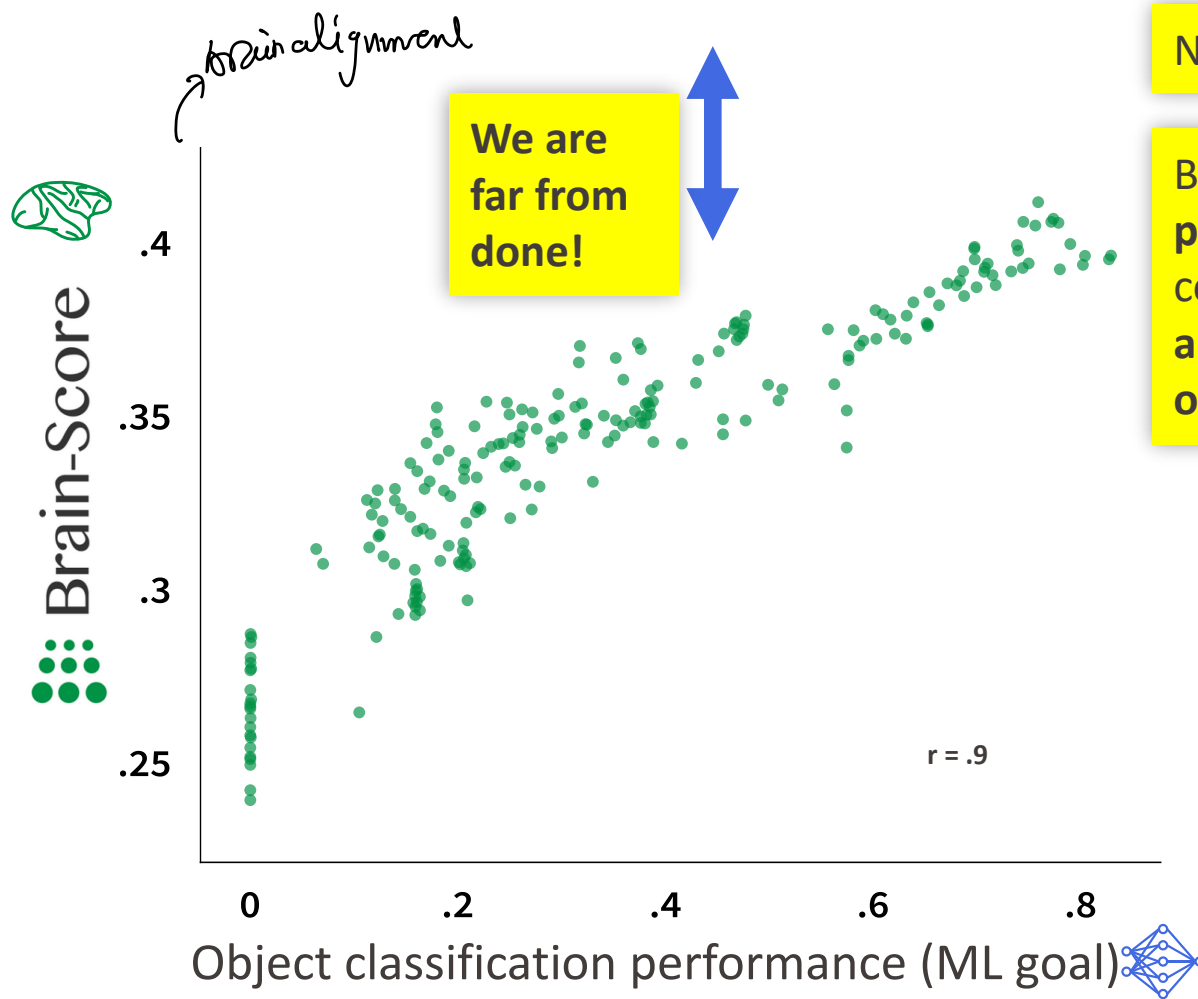


■

Normative variable

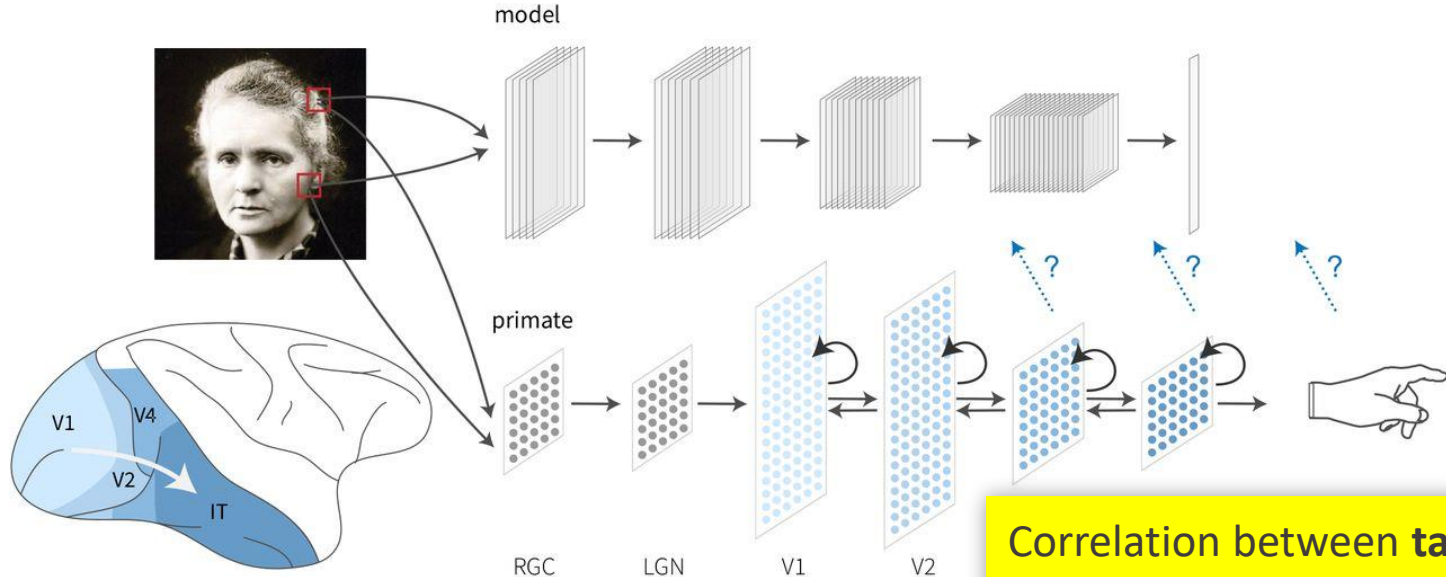
cf. Yamins, Hong*, et al. 2014*
Schrimpf, Kubilius*, et al. 2018*

EPFL What explains the model differences?

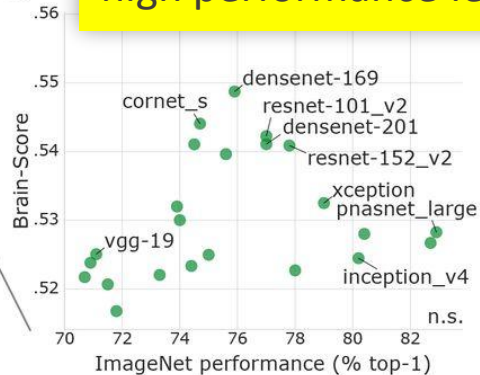
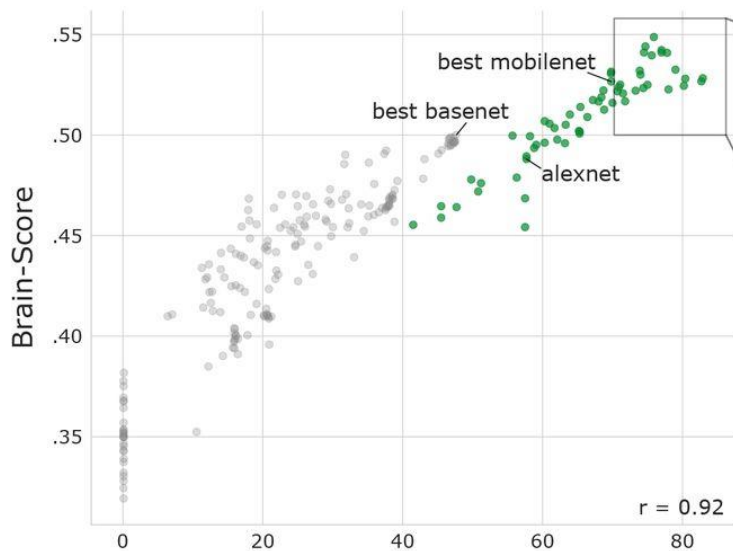


Not the small-scale circuits

But rather **task performance**, as a combination of **architecture + optimization**

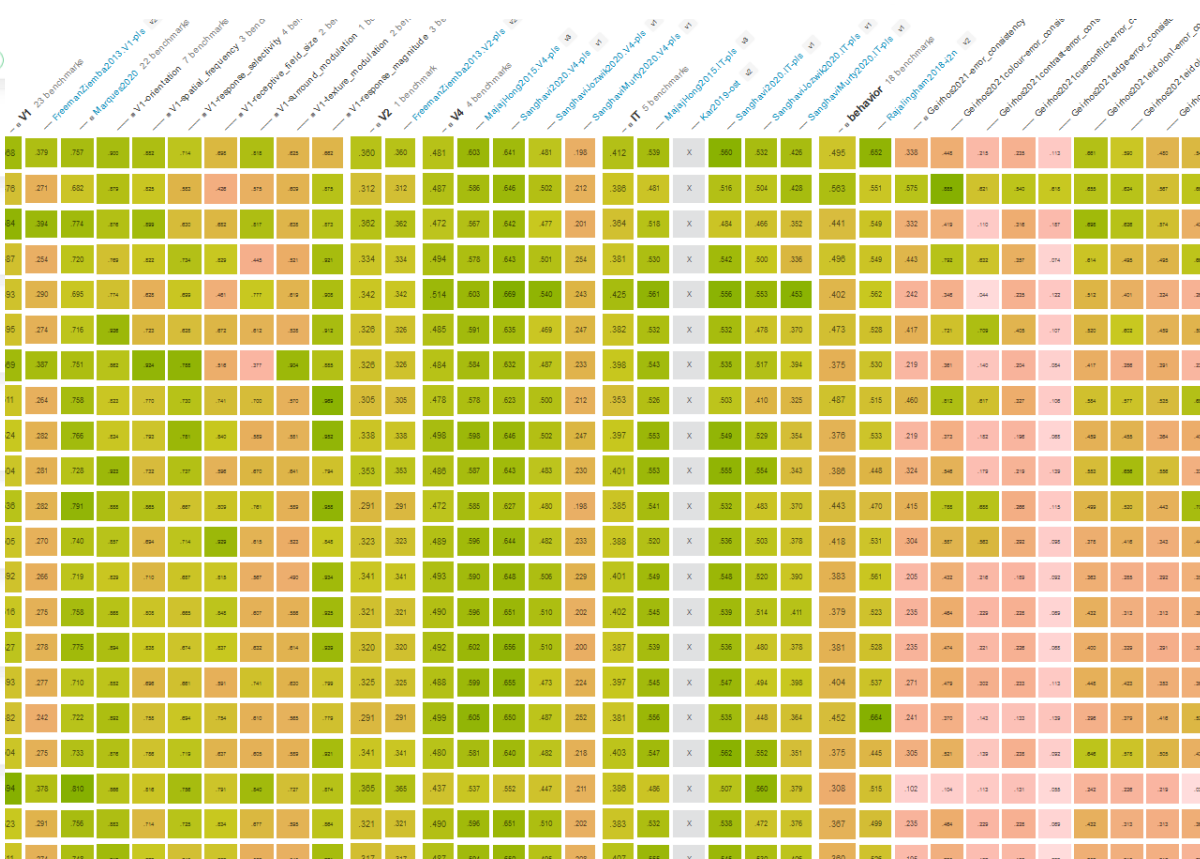


Correlation between **task performance** and **brain alignment** disappears at very high performance levels

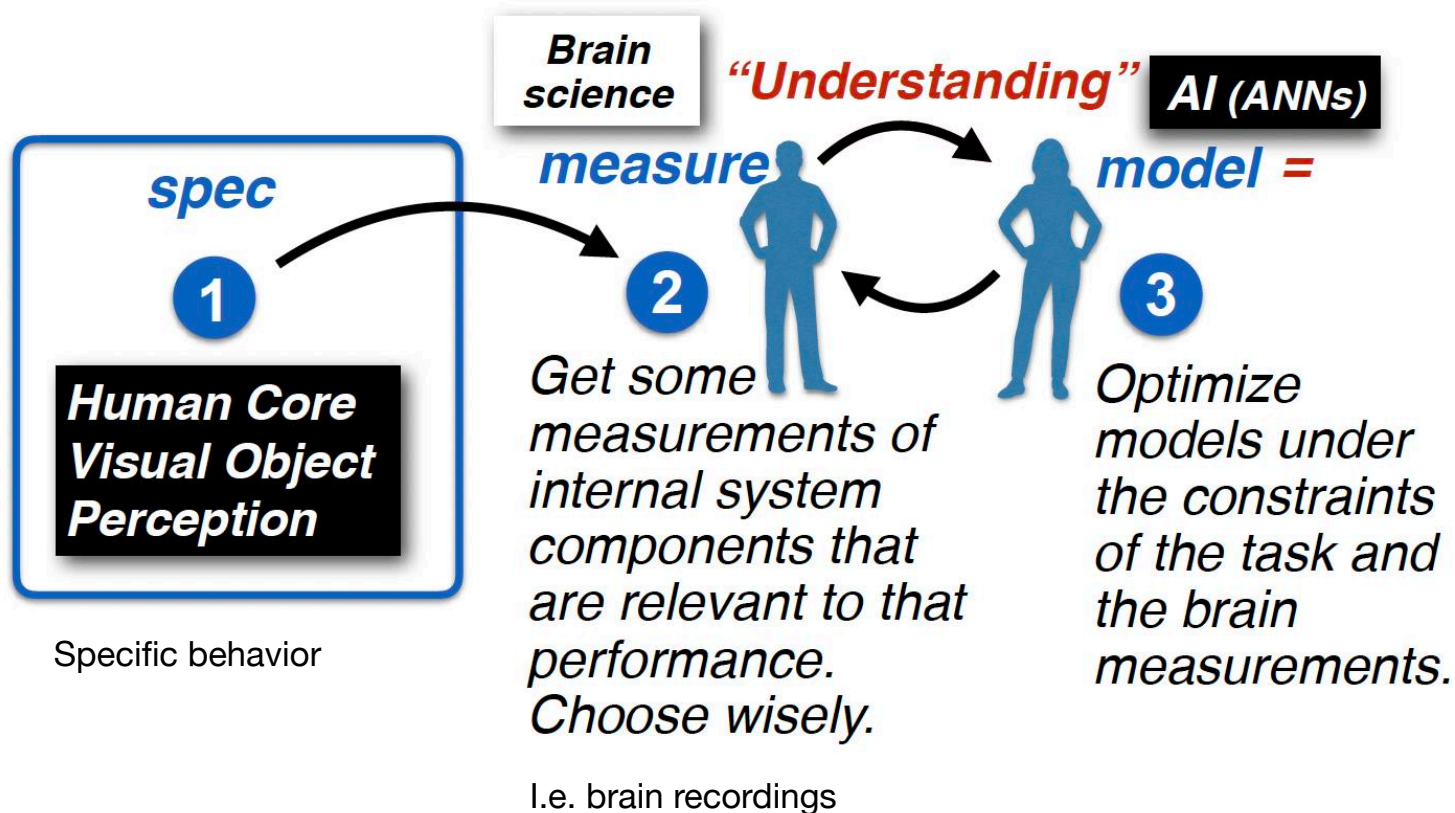


Good performance, but brain alignment is still too low. But it is too important to increase that, because we are trying to reproduce the brain!!!

Rank	Model	average 81%	neural 38%	behavior 43%
1	convnext_large_mlp:clip_laio	.467	.378	.555
2	convnext_xlarge:fb_in22k_ft	.449	.338	.560
3	vit_base_patch16_clip_224:o	.445	.343	.548
4	vit_large_patch14_clip_224:le	.445	.332	.559
5	vit_large_patch14_clip_224:o	.443	.341	.545
6	vit_base_patch16_clip_224:o	.442	.352	.532
7	vit_relpos_base_patch16_cls	.437	.381	.494
8	cvt_cvt-w24-384-in22k_finetuned-in1k_4	.430	.327	.533



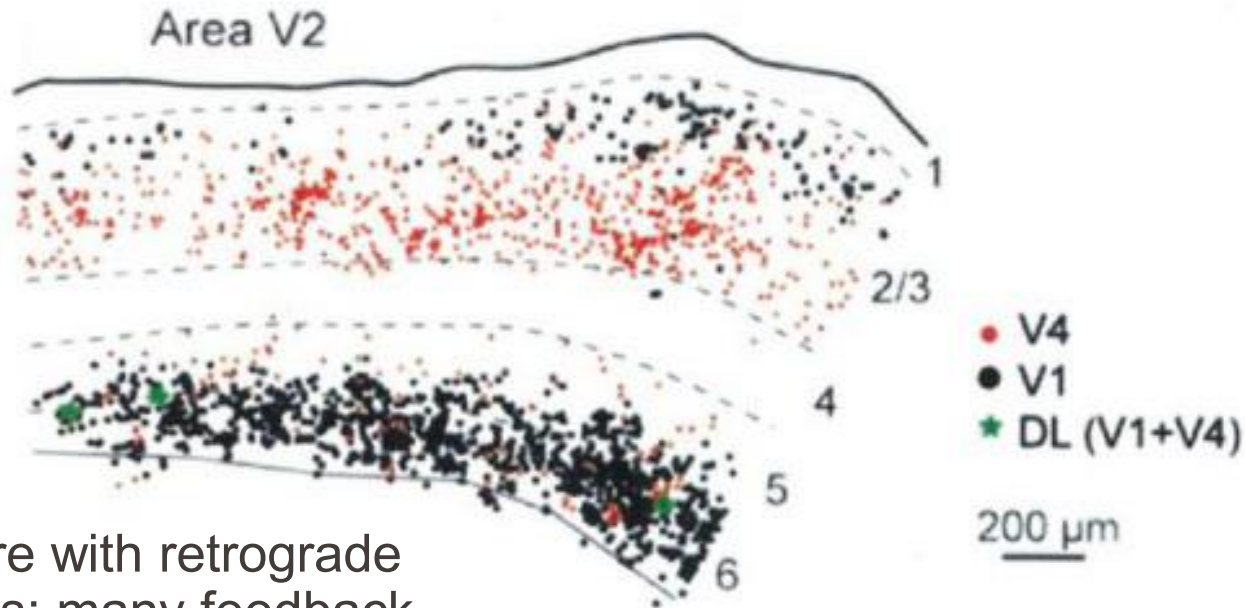
The reverse engineering principle



This is what im doing!!

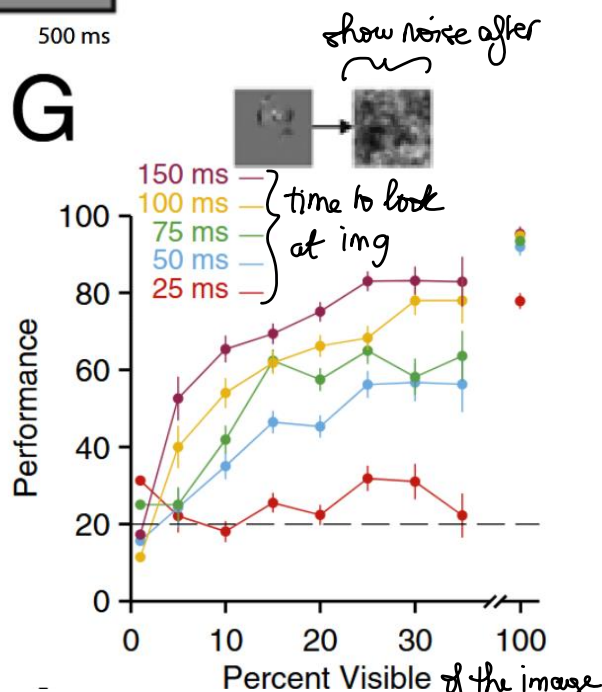
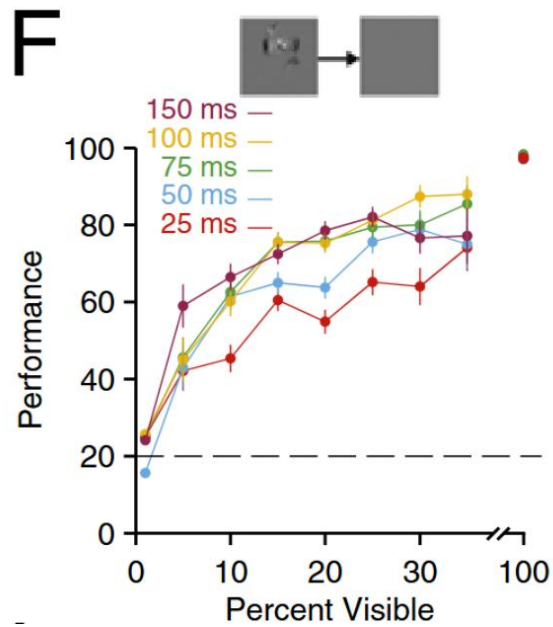
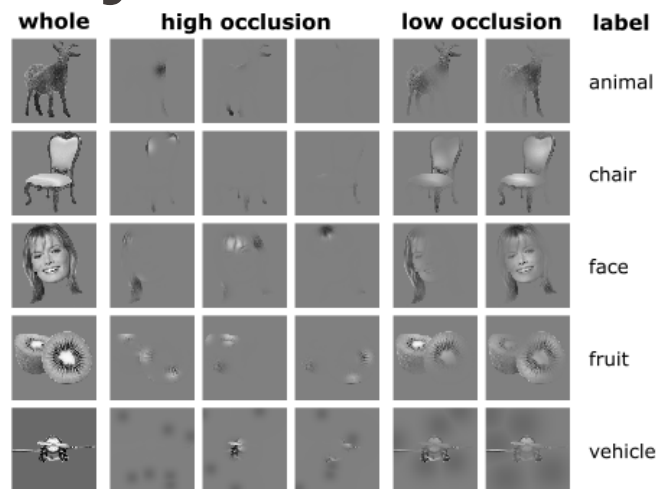
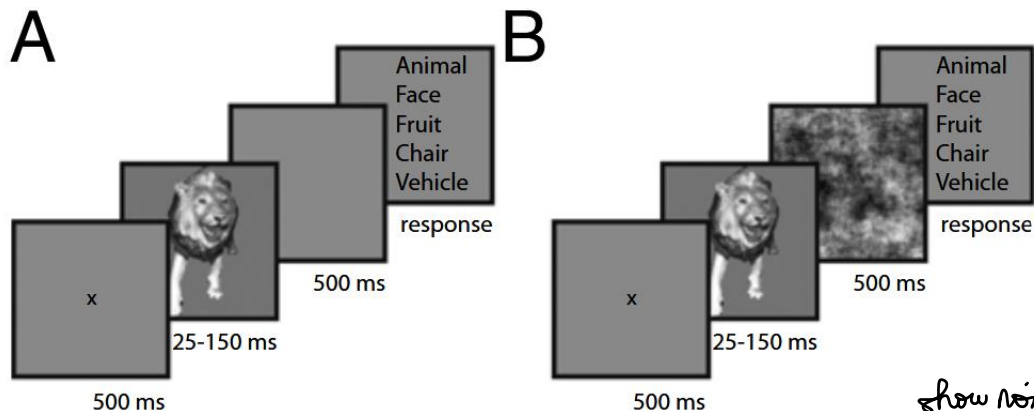
Feed-forward and recurrent processing

- So far, we have considered the visual system as a hierarchical **feed-forward** feature encoder
- But: many connections in the brain are **recurrent** (depending on the area thought to make up even 80% of all connections)



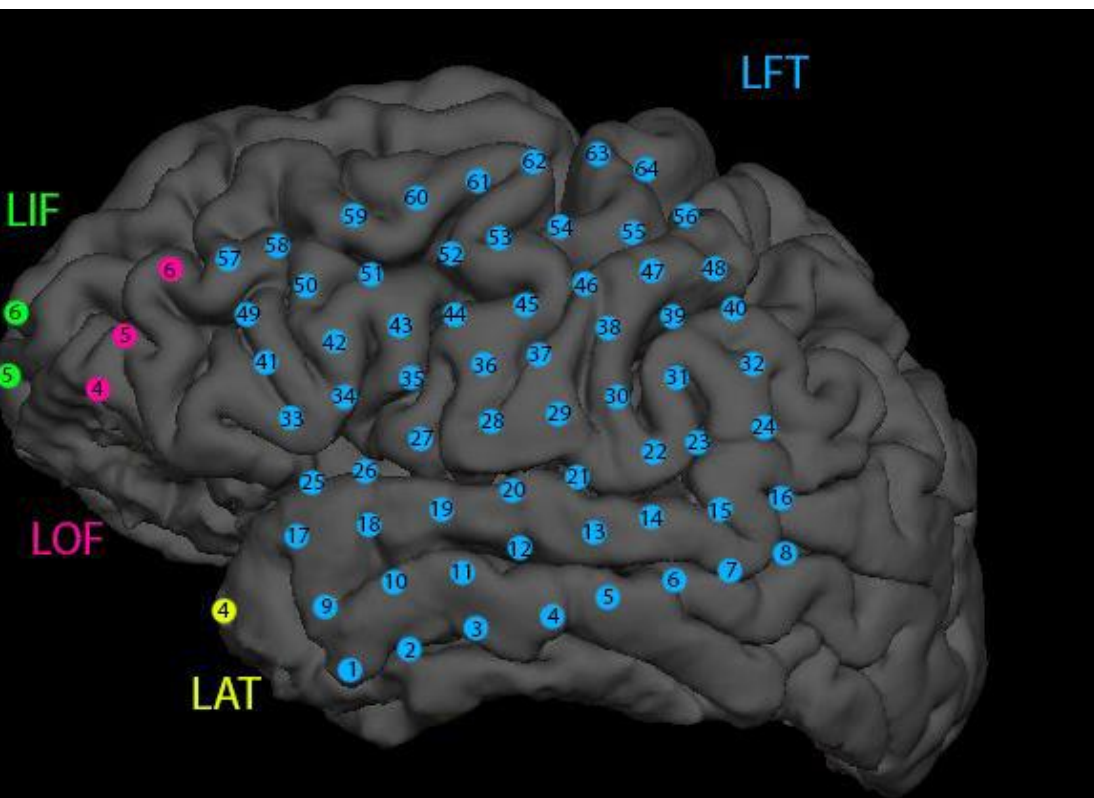
E.g., here with retrograde injections: many feedback connections from V4 onto V2

Recurrent processing in the visual system

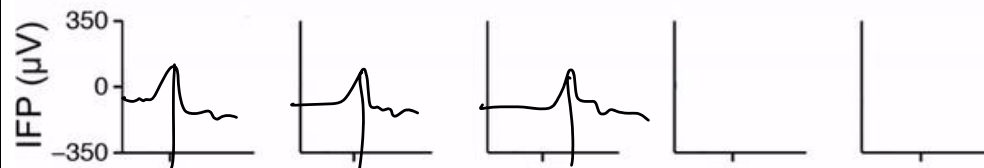


- Humans can classify images better the more visible they are
- Backward mask: thought to disrupt recurrent processing
- With backward masking, performance drops at short presentation times

Recurrent processing in the visual system

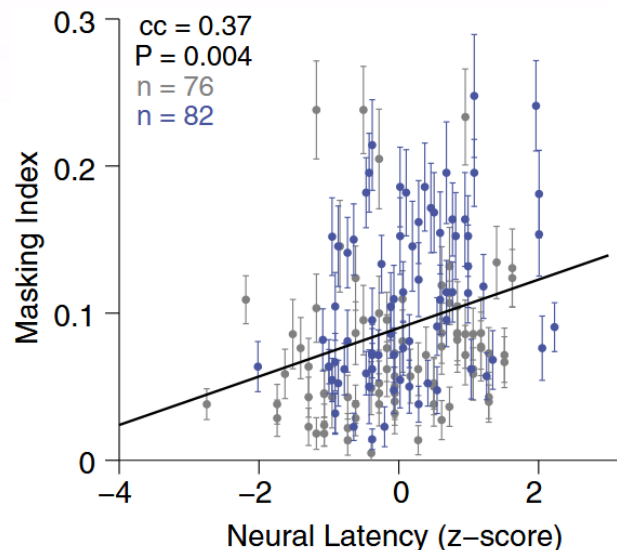


Electrode in left fusiform gyrus (face-selective)



Time (ms)
peak comes later when you see less of the img (latency)

Higher neural latency during occluded image presentation
 → stronger effect of masking on behavior



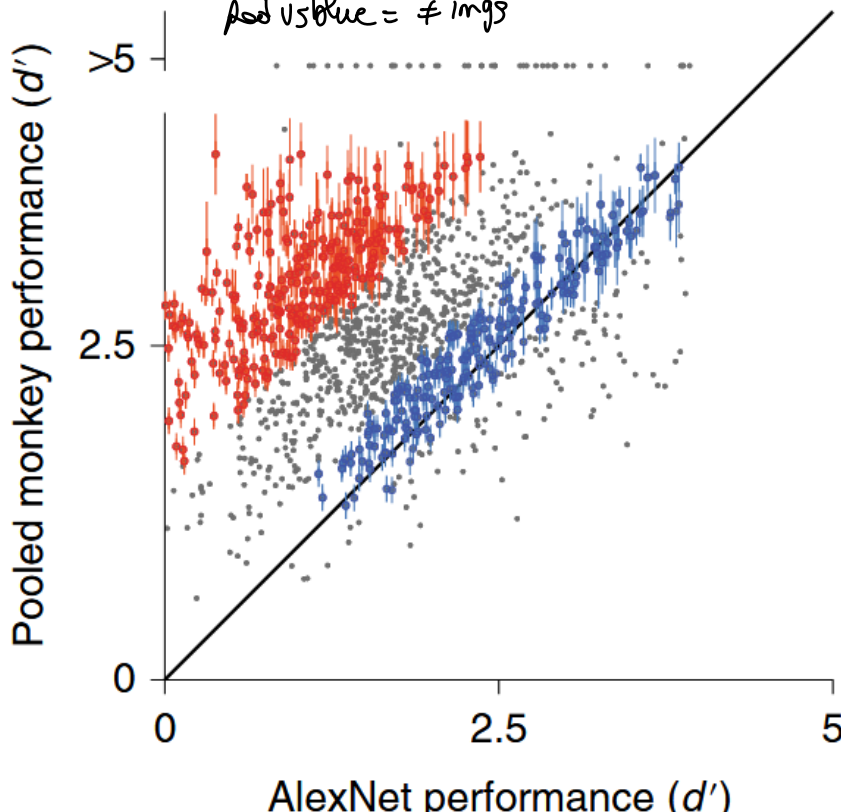
Tang & Schrimpf & Lotter et al. 2018

Recurrent processing in primate IT

- Model-guided stimulus selection: choose images where a feed-forward model struggles, but humans perform well.

Behavioral comparison
monkeys vs DCNN

red vs blue = diff imgs



Object discrimination task

Monkey perform better than model on challenge images. Why?

Fixate
300 ms

Test image
100 ms, 8°

Blank
100 ms

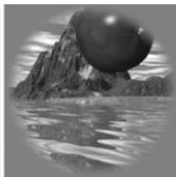
Object choices
1,500 ms

Trial time

Challenge images

Dog

Apple



Elephant

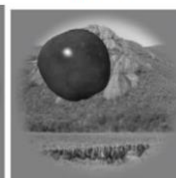
Chair



Control images

Dog

Apple



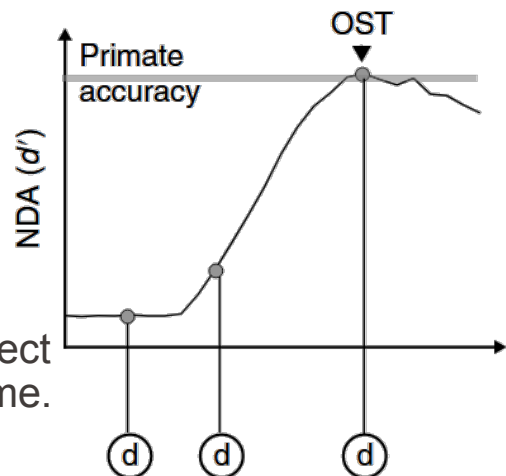
Elephant

Chair

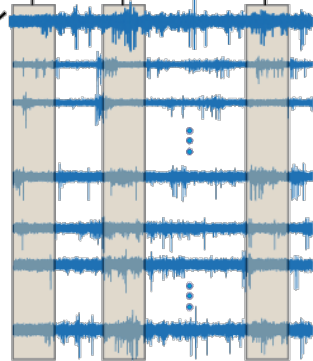
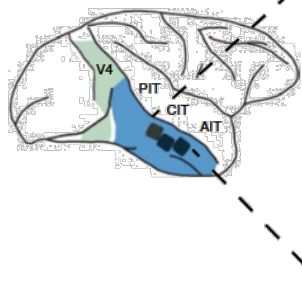


Recurrent processing in primate IT

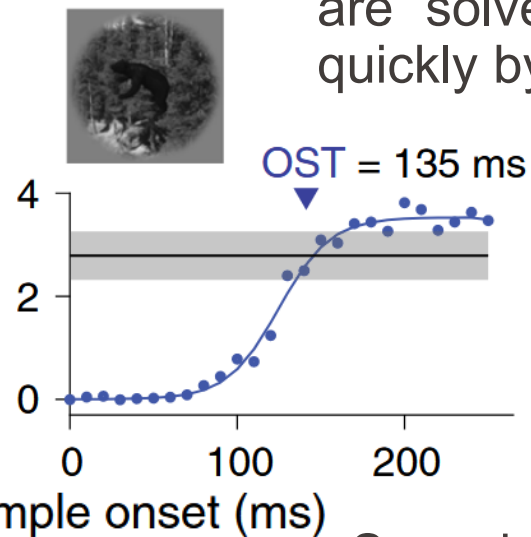
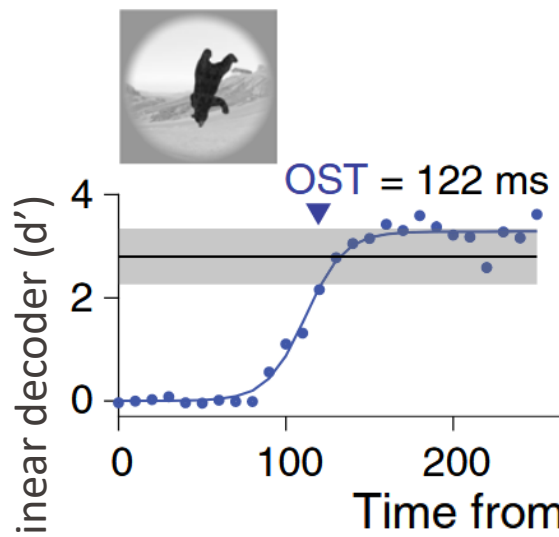
- Some images are “solved” quickly by IT



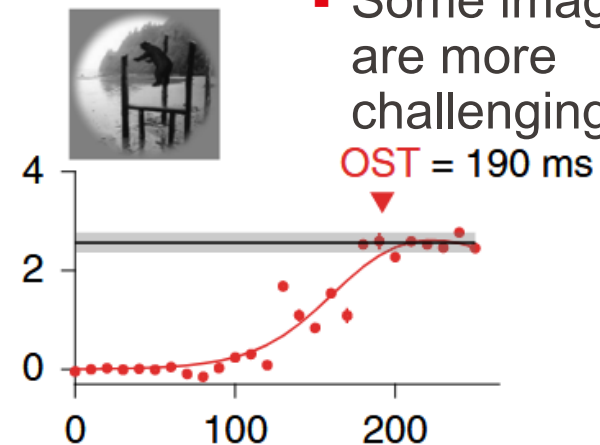
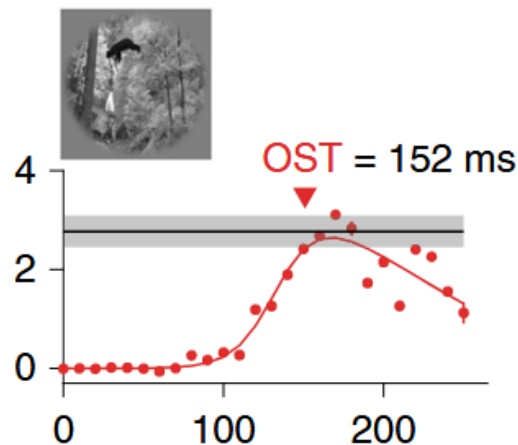
- OST = object solution time. When are categories decodable from IT



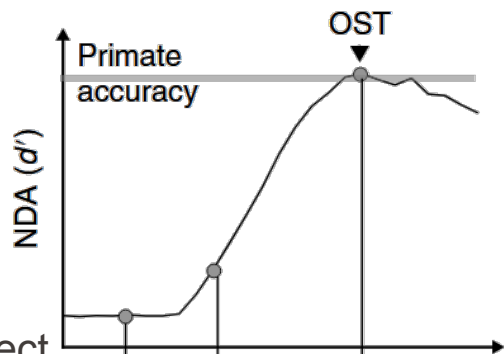
Time from sample onset (ms)



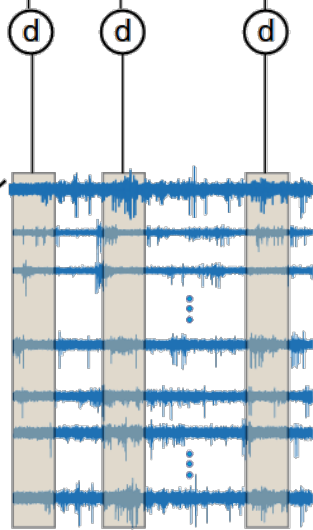
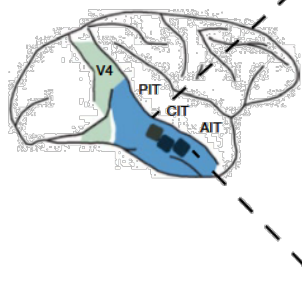
- Some images are more challenging



Recurrent processing in primate IT



- OST = object solution time. When are categories decodable from IT



10 ms

Time from sample onset (ms)

- Control images are solved quickly

- Challenge images require more processing



"face"



"zebra"

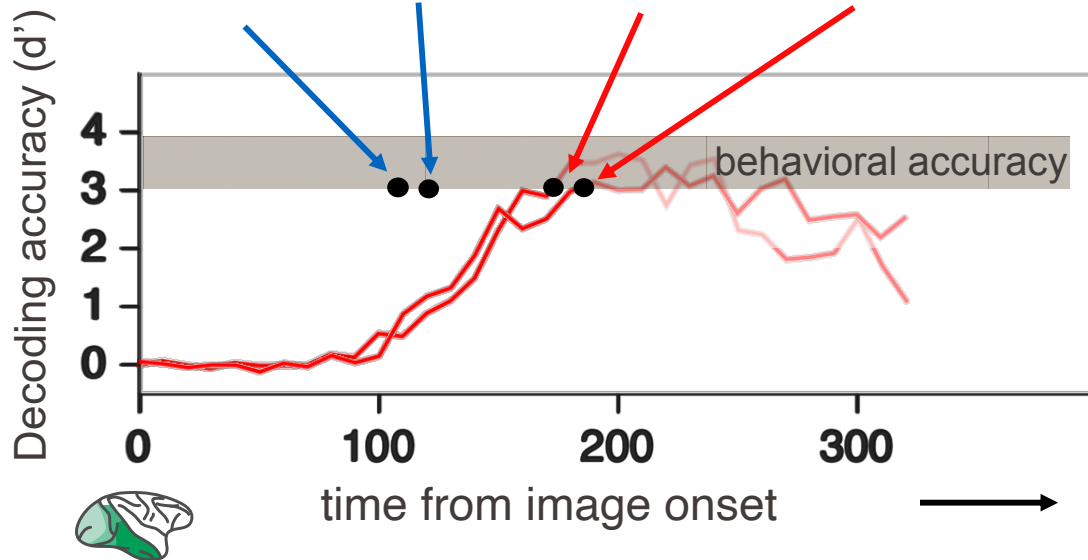


"car"

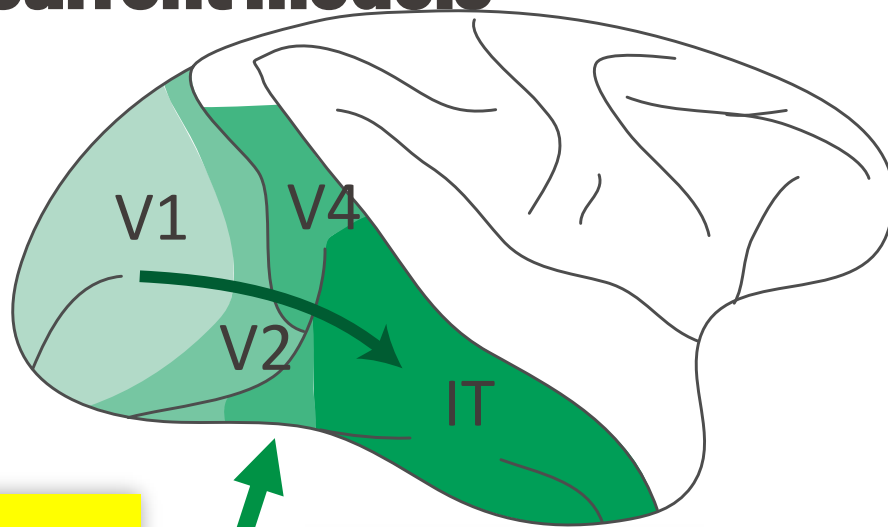


"dog"

...



Modeling recurrence: transform feed-forward networks into recurrent models



1. difficult to map anatomy

2. no temporal responses

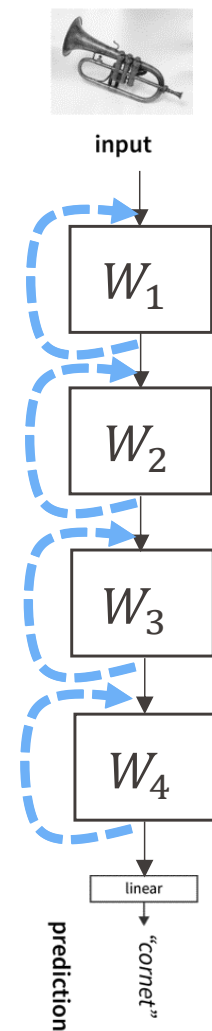
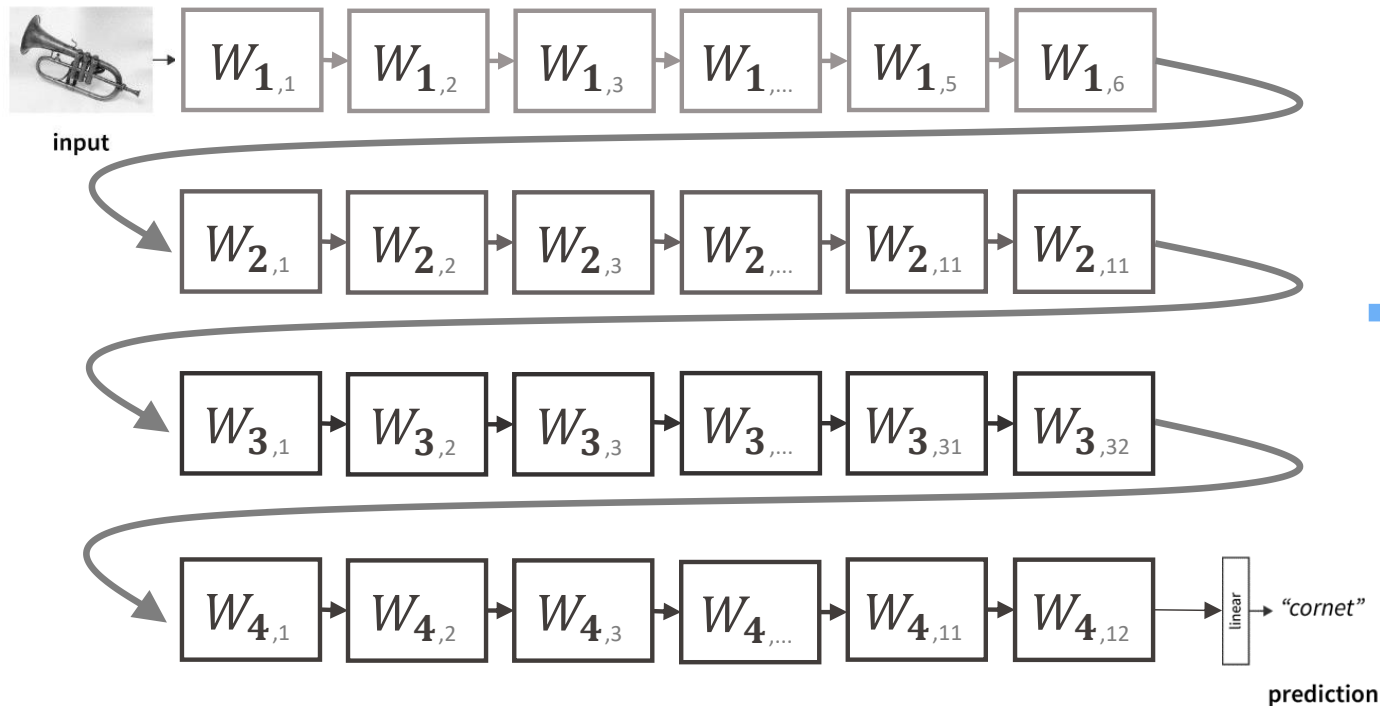
e.g. ResNet-101

V4?

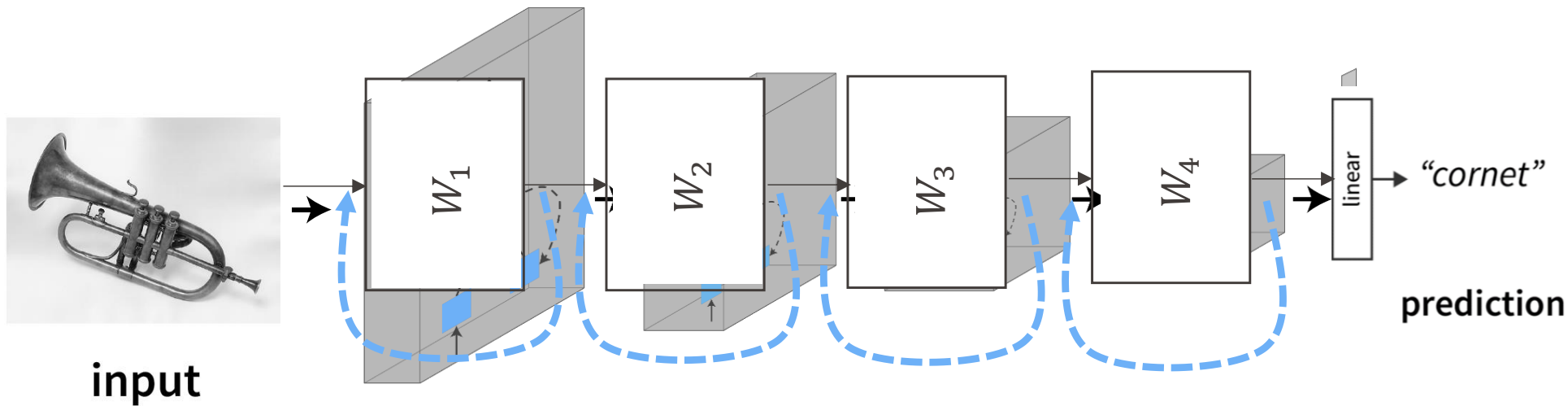
IT?



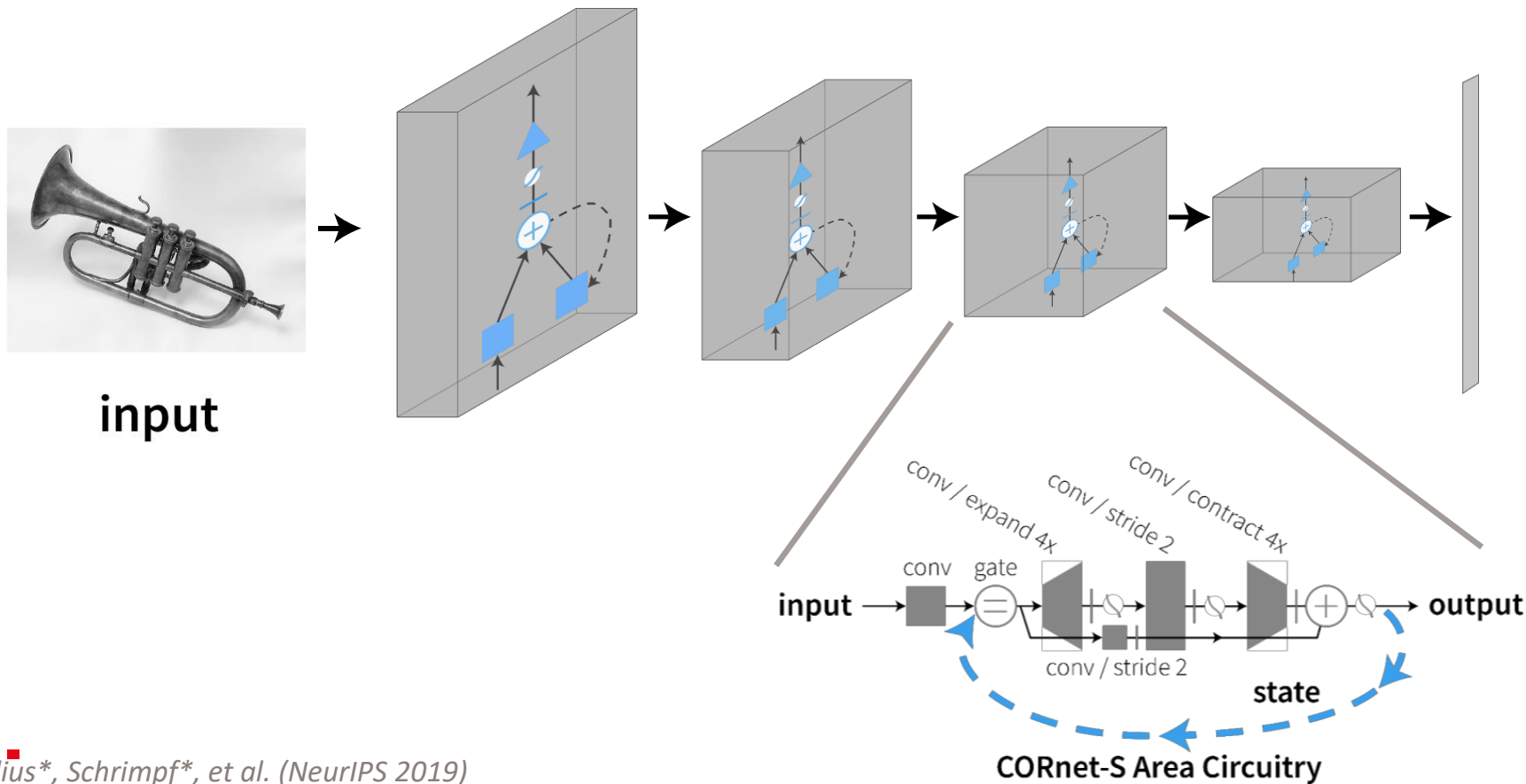
Transform feed-forward networks into recurrent models



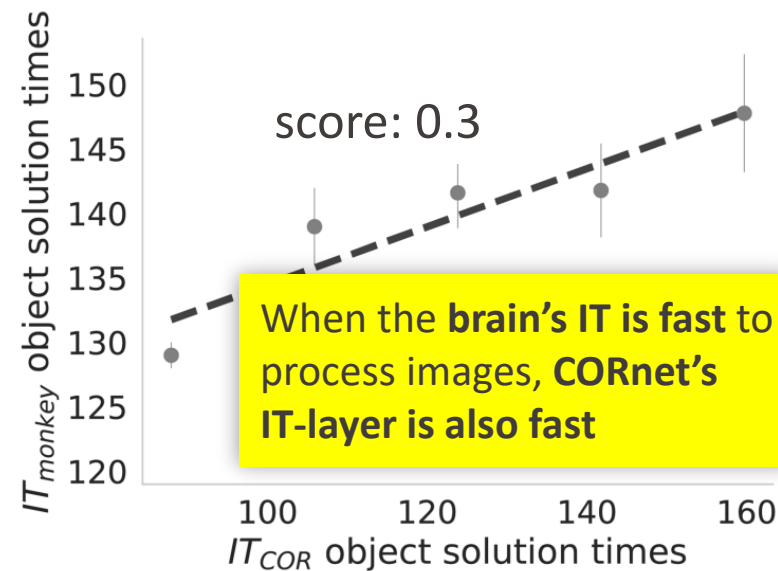
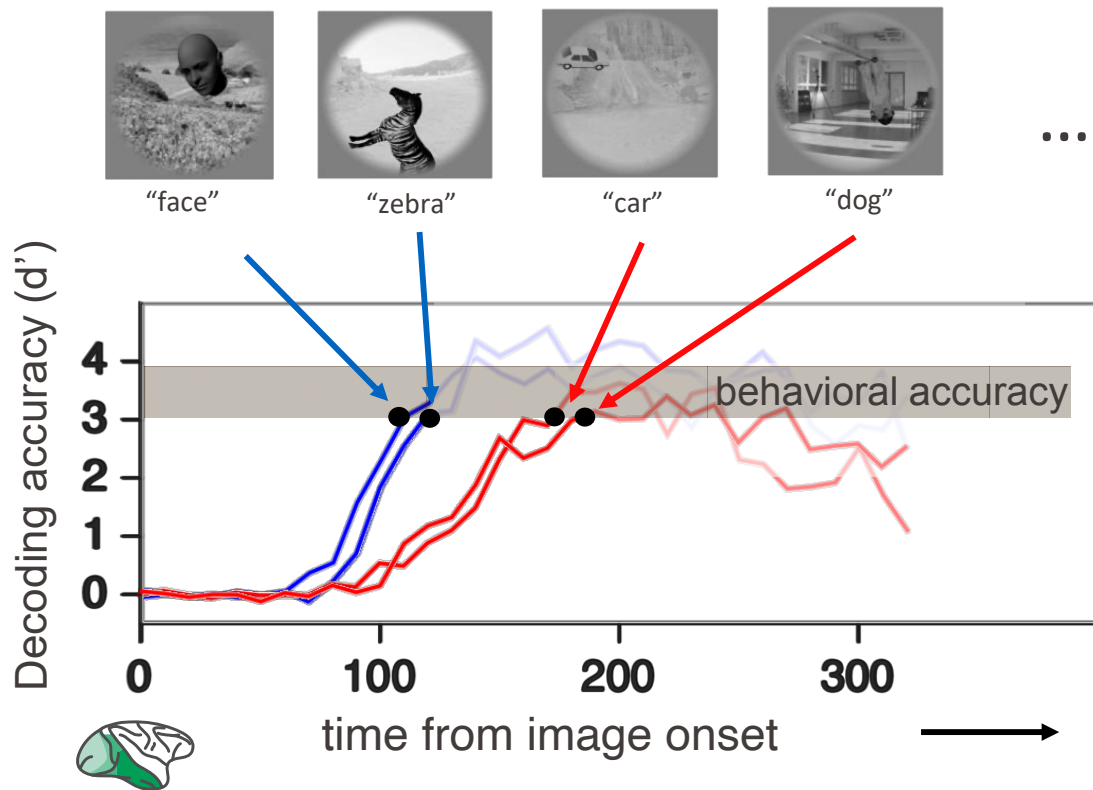
Modeling recurrence: transform feed-forward networks into recurrent models: CORnet



Recurrent CORnet model: compact architecture via recurrence



Recurrent model predicts temporal dynamics in IT



Unlike feedforward models, CORnet-S can **predict neural responses over time**.

Take-home messages

- Reverse engineering principle
- Representation learning with goal-driven / task-driven learning provides a normative framework for studying the nervous system.
- Models trained on object-recognition can predict neural activity in ventral pathway during object recognition well. Outperforming previous models by a large margin.
- The work by Yamins, Hong, DiCarlo and collaborators highly influenced vision research and inspired much other work beyond vision.
- Recurrent processing is recruited for more challenging images. Compact recurrent models explain the brain's temporal dynamics to a first extent.
- In the exercises you'll recap linear probing & explained variance. Next you'll start a mini-project modeling IT neurons!